

Identification of a Personalized eGFR Threshold Improves the Prediction of Kidney Failure Risk After Transplantation.

Corresponding Author: Dr Haralampos Hatzikirou

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In this paper, the authors propose an approach to modeling the temporal evolution of kidney graft failure that combines a dynamic GFR mathematical model and machine learning that also accounts for parameter uncertainty. This strategy also includes a way to determine individualized predictions of the GFR threshold for specific patients. The results suggest that this approach achieves greater than 80% accuracy for the first and second year after kidney transplantation. Overall, this work is a very interesting combination of mathematical modeling and machine learning optimization that is valuable for the field to consider as a methodological option. This manuscript is largely clearly written, however detailed in some specific comments below, this work could benefit from adding a number of additional details, including further specifics on the machine learning and machine learning evaluation elements of this work and context to more clearly label or explain some figures in the paper, to make this paper clearer for a broader audience. Additionally, I do have some key concerns regarding some of the results presented in this paper, and how these translate to conclusions presented. While some/all of these may be addressable with additional details and clarifications in the manuscript, others may require additional analyses to be convincing. Below, are specific 'major' and 'minor' comments.

Major:

- Fig. 1 – the distributions look the same or very similar, with almost entirely overlapping interquartile range. How are there significant differences?
- The results in 2.4 are unclearly presented with respect to comparing the proposed method to XGBoost and Random Forest since those ML algorithm results are not broken down in the same way as the proposed methods as illustrated in Fig 3. From the description it seemed like the authors proposed method benefitted from additional tuning to the data, than would have been available for the two ML algorithms used. Furthermore, there is no mention of the authors applying a hyperparameter sweep for XGBoost or Random Forest to give these algorithms the best chance of success (standard practice for ML modeling). Presumably default run parameters were used for both ML algorithms, but additional details about how these algorithms were applied (e.g. what hyperparameter settings, and clarify any differences in how the proposed method in this paper vs. these two ML algorithms are being applied as a GF classifier).
- Applying cross validation vs. a single partitioning of the original data into a training and validation cohort, since the results are open to sample bias (i.e. the instances in the validation set could be easier or harder to predict accurately just by chance).
- Linked to the above comment: -How were the training and validation cohorts determined? Did you start with a single dataset and randomly partition the instances? Did they represent two patient groups collected over different time intervals? These details would be necessary to consider other potential limitations of the presented analysis given that only a single data partition was utilized.
- Is the accuracy in this study based on a binary class outcome i.e. rejection vs. no rejection, or multiclass, i.e. no rejection, one rejection, multiple rejections? This is not always clear in the paper from section to section and figure to figure. Either way it seems like the target outcomes is imbalanced (as mentioned in the supplemental materials), meaning that standard accuracy, and AUC of ROC may not be the fairest evaluation criteria, due to the class imbalance. Authors should consider balanced accuracy and AUC-PRC as an alternative or in addition to ROC-AUC.
- From section 4.3, it seems possible that interpolation of GFR values could impact the presented results. Can the authors include discussion on this consideration? Are there any strategies that could be employed to account for this possibility (or avoid them, using the data as is without interpolation)?

-It's not clear how generalizable this solution would be. Would this method need to be adapted to different patient populations, e.g. when applied at different clinical institutions. Do the authors expect or have plans to evaluate this model on other patient samples, and larger sample sizes? If so would this algorithm need to be retrained or reoptimized for different populations?

Minor:

Line 99 – reference error.

-Supplemental materials are somewhat difficult to follow in the context of the provided main paper text and additional supplemental materials text. I'd suggest providing more context and explanation in the supplemental materials if there is no space in the main text.

-Fig 2C, not enough context is provided in the paper to explain what this figure depicts clearly. I'd check all figure and table legends and update them as needed to provide greater context as needed.

-I found it a little confusing to re-name the training and validation datasets as N1 and N2 groups. Linked to this, on line 343, the authors mention they validate their predictions in two independent datasets, but only one validation dataset was apparently described and utilized.

Section 3 mentions the use of state of the art feature selection techniques. What makes the ones used, 'state-of the art', since they seemed fairly traditional based on my experience.

-Figure 5 is a bit distracting with the black background and white text. This figure could be more clearly constructed overall.

Reviewer #2

(Remarks to the Author)

There is no doubt that kidney transplantation (KT) is the best treatment for patients with advanced CKD (preemptive) or ESKD. However, we are limited by the number of available organs and the longevity of the allograft.

I commend the authors of this paper for attempting to apply a mathematical model and machine learning capabilities to try and answer an important question of 'which kidney allograft will last longer?' using the most available measurements - GFR, and thus coming up with a model to calculate individual graft failure risk.

However, I have a few concerns regarding this work.

First of all, their goal is to use 2 early GFR measurements in KT recipient (KTRs) to predict the function of the graft at 1 and 2 years. In this era, the short term survival of allografts is believed to be excellent, with 1 year survival of 98% and 3 years of 95%. Allograft dysfunction and failure often happen later on (ie in the 2nd decade of the allograft life), and that's where we need to improve outcomes. Year 1 and 2 allograft function - are not necessarily indicative of what happens long term with longer exposure to CNIs, later rejections, weight gain, worse DM control, etc.

Moreover, this study spans over 15 years for KTRs, in which kidney allograft survival has significantly improved with changes to the medications, surgical techniques, and so function from the authors testing or validation cohorts for early 2000s does not apply now. The authors do mention the change from CYA to Tac during their study period, but no details are given to how many pts were in each group.

There is also a lot of clinical data missing, such as levels for CNIs in the timeframe of the GFR tests, which are usually intentionally higher in the first 3 months (where their first and sometime 2nd calculations is from), and can cause the GFR to go down.

Additionally, their groups are divided based on rejection yes/no/multiple in the first 6 months after the txp (I am assuming, as the time for the rejection "cut off" is not clear), but we know that earlier rejections are often treatable and are not necessarily associated with later function. Also - while the authors so mention types of rejection and using the Banff score for grading, there is no data of type and grade of rejections in the cohort (specifically rate of each type), which makes it harder to extrapolate - treated early TCMR Banff 1a outcomes are not the same as TCMR grade 2 or ABMR.

Additional comments -

1. Line 352+353 notes that these cohorts included KT alone and KTs combined with other SOT in the methods section. These groups do not behave the same, and there is only one mention later on of KT vs SPK, which also differ as at least in the US - SPK organs are often of better quality. But, I am also left with the question of whether or not there were other SOT but pancreas included here, and how many?
2. The initially refer to a training group (N1) and validation group (N2) as having similar number of patient (around 850 per group), but only later do they note that they included only the pts with NO rejection (N=372, hence less than half the group) in the training cohort for the model.
3. Missing data of BK - specifically if BK was seen on bx (BK nephropathy), as a cause of GFR decline.
4. Lines 333-334 their assumption of "undisturbed transplant course" = patients with no rejection is simply wrong. There are many factors besides rejection that are not addressed here and impact the graft function, such as surgical complications, AKI, hypotension, CNI toxicity, infections (UTI, BK, and more). Not using them in the model, weakens its strength.
5. Patients in the validation group had notably shorter CIT than the training group. CIT is known to be a factor of allograft outcomes.
6. There is no data re race and ethnicity in the study, and we know that race (of donor and recipient) impacts allograft function in KT.
7. There are several statements in the beginning of the manuscript - line 47-48 and 48-49, for example, that are too definitive and not necessarily true based on available data. An example is "In particular, the GFR is determined by the quality of transplanted organ", were in fact the quality of the organ is an important but not the sole determinant of graft function.

2 smaller comments -

- I would use the terms chronic kidney disease and kidney failure as opposed to "chronic renal disease" and "renal failure" to make the paper more uniformed (use of kidney in the context of kidney transplant is used throughout)
- line 256 I believe instead of "donation" it should be "transplantation" - as this paper is about the recipients.

I acknowledge that some of my issues with this manuscript/ study were listed as limitations by the authors, but I do not believe that negates the importance of these comments for this work to be significant clinically and therefore applicable to the KTR patient population.

Reviewer #3

(Remarks to the Author)

The authors aim to predict the graft function dynamic based on pre-transplant data and early post-operative graft function.

The state of the art is missing. There is number of predictive tools in kidney transplantation aiming to predict the graft function and untimely the graft failure. The author must consider this literature in the introduction, better explain the originality of their proposal, and finally report the increase of prognostic capacities related to their approach.

In a methodological point of view, the understanding of the modelling is complex. It appears as a home-made methodology which was not validate elsewhere. I did not identify how the authors properly deal with censoring, repeated measurements, and joint modelling of the time-to-event.

In conclusion, I did not believe that this work will improve the existing literature.

Reviewer #4

(Remarks to the Author)

Nature Communications Review

The paper the authors present offers a prediction of kidney failure risk after transplantation based on a multi-faceted approach which includes classic regression techniques and dynamical systems modeling. Their approach is very interesting and warrants consideration, however as a reader I often struggled to understand their methodology and their justifications for the various approximations and conceptual leaps they made between different steps in their analysis. Specific comments are below:

1. Lines 113-120 – you performed multi-linear regression with the non-rejection patients as your training set, and your results show that the R^2 for the rejection patients is around ~ 0.5 . Why not include regressions for the rejection patients as your training set? It seems that your analysis will thus be mostly valid for the non-rejection patients, as $R^2 \sim 0.5$ is not very high. If you include the rejection patients in your regression, perhaps there will be more than 3 major features as you found and the regression will better capture all 3 categories of patients. In short, isn't clear to me your choice of regressing only on non-rejection patients. Same for lines 406-408 – you show that the assumption isn't very good.
 - a. Lines 37-39 in supp. material – you should reverse the order in which you state the features. According to Table S1, the most important feature is GFR_in, while in the text it seems like the donor's age is most important.
 - b. Lines 59-60 in supp. Material – R^2 decreases and MSE increases. You write that both decrease.
2. Lines 174-179 – A quantitative test to check the goodness of fit of the sigmoidal regression is missing. Furthermore, how were the survival probabilities from the Scheffner et al paper calculated? It isn't clear what the y axis actually measured in Figures S5-8. Is survival probability itself a result of regression, or is it measurable? It is strange to see %s of survival at 15 years following transplantation without an explanation of how this survival was estimated. Were there additional inputs in the sigmoidal regression in addition to GFR365d in the original Scheffner analysis? What would happen if you included them in your regression?
3. 4.1 Approach Rationale – I strongly suggest this should feature in the introduction as it explains the approach of the whole paper. It should be among the first things someone reads when opening the paper.
4. General comment – in the mathematical modeling section, the authors sometimes use "a" and sometimes "" interchangeably. See for example S.2 vs S.8. This happens often, should be consistent.
5. Lines 120-121 in supp. Material – why do you say that alpha and gamma are small? In your derivation you didn't supply numerical values till now. Also, small compared to what? They have different units, so explain what exactly is small and why. You can also solve analytically for gamma from S.13, define $\gamma = t^2$ and solve for t. Then take square.
6. Lines 146-148 in supp. Material – in this section you take many approximations/assumptions that aren't explained.
 - a. Why do you assume that $du/dt = 0$? How is this justified.
 - b. Assuming $du/dt = 0$, you now take $u = -2.07$ constant for all, and then take $\alpha = 0$. Why? Earlier you said that alpha must be positive (line 115). Why is u constant for all, and if you take $\alpha = b = 0$, then you don't have a double-well potential anymore but a simple damped harmonic oscillator. This jump isn't clear to me at all.
 - c. Then you assume that $\lambda = 1$ and get an estimate for γ . However, here also, it isn't clear why you assume that $\lambda = 1$. You then use this value of gamma for all patients, however, it isn't clear at all why it would hold for different patients considering the unjustified assumptions you took along the way.
 - d. By assuming $\lambda = 1$ you got an estimate for γ . You then keep γ constant while drawing λ s for different patients. The explanation for this inconsistency is not explained anywhere in the paper.
7. Lines 331-333 – you write that the first-year FGR value is a good proxy of the functional graft performance, however based on the sigmoidal regressions you showed in S5-8, it isn't entirely convincing.
8. Fig 5. Should be much clearer. Which model parameters are estimated using GFR_365 days? Which critical thresholds

and uncertain parameters are estimated? The uncertainty analysis step also isn't clear.

9. Is Fig.6 real GFR data? Where is it taken from?

10. Lines 462-466 – it isn't clear at all how λ was estimated for each individual. Especially, as stated above, you use an assumption that $\lambda=1$ to get the numerical value of γ . So how are you now estimating it? How did you reach the distribution in Fig. 5? Also, you show in that figure that most values of λ fall within [0-0.2], so how could you assume that $\lambda=1$, use it to find γ , and then use the value of γ to give a much lower value of λ ? The whole treatment of λ is very unclear.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Overall I'm satisfied by the edits and authors responses to reviews, from the perspective of a reviewer with Machine Learning expertise. I would note that it might be made clearer on page 4 that while cross validation was applied to multilinear regression it was not applied to the later part, perhaps with some additional context provided for clarity as to why.

Also in the discussion section I would strongly encourage the authors to lay out any potential advantages their approach may have had (with respect to differences in how the analysis was conducted) in contrast with the other ML analyses, and how this may have impacted the performance differences observed (even if it alludes to future work that should be examined).

Reviewer #3

(Remarks to the Author)

The state of then art in nephrology and robust modeling longitudinal process jointly with time-to-event are still missing. The authors seemed to propose an home-made methodology. They have to demonstrate its performance against the state of the art, for instance by simulations. Such an application does not allow us to validate the proposal. I believe that this paper should not be considered.

Reviewer #4

(Remarks to the Author)

I thank the authors for their revisions and clarifications regarding the Duffing oscillator–based approach for GFR evolution. The updated manuscript appears considerably improved. I have several specific comments regarding the manuscript's treatment of the Duffing oscillator model for GFR evolution:

The handling of γ (gamma):

Is a constant γ value used across all patients in the simulations? Does Table S4 only include data from patients without rejection? If the γ value from non-rejection patients is applied to model potential rejection cases, what is the justification? Has sensitivity analysis been performed to demonstrate result robustness across different γ values? What is the expected biological variation in γ between rejection/non-rejection cases? How critical is γ to the model outcomes?

The revised treatment of λ (lambda) is now more rigorous and well-justified.

Visualization Recommendation

I strongly recommend adding a comprehensive schematic, graphic figure (ideally as Figure 1) that illustrates:

The complete patient workflow

Step-by-step methodology

Key decision points and processes

Example patient progression through the model

This visual pipeline would significantly enhance reader understanding of the methodology and make the paper more accessible to a broader audience. This figure should illustrate the actual steps as currently written in Figure 1.

The authors have made substantial improvements in addressing previous concerns about modeling assumptions (particularly regarding du/dt , α , b , and λ). However, the above points should be addressed to further strengthen the manuscript.

Reviewer #5

(Remarks to the Author)

In this manuscript, the authors attempt to prove a personalised eGFR threshold to detect post-transplant kidney failure risk.

1. You mention the analysis indicates that an eGFR threshold exists that allows for classifying high-risk patients. I struggle to

accept there is a sharp dichotomised eGFR threshold for risk above or below with risk for kidney failure risk dramatically changes. eGFR and risk for kidney transplant failure is a continuous variable and influence by many other variables such as proteinuria. I cannot see the latter factored in at all? A high eGFR but high proteinuria would not reassure me of low risk for graft loss.

2. On page 14 you mention the generalizability of your approach with sensitivity analysis stratifying both the training and validation cohorts based on patient sex, donor sex, and patient age. Why did you choose these variables - especially donor sex? That seems an odd choice to show generalisability. Generalisability should be in a different ethnic groups, different immunosuppression regimens etc.

3. I struggle to understand the clinical application of this work. Graft function and the performance of the kidney allograft is based on many donor, recipient and surgical factors and numerous post-transplant variables.

4. I am not sure why there is such focus on stratification by rejection but nothing for other causes for graft dysfunction.

5. How would this compare with tools like iBox? What would be its advantages or disadvantages?

Version 2:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

No relevant modification in the paper. I still believe that this paper must be rejected since it does not use robust modeling and it will have no impact for both the communities in nephrology and mathematics/statistics.

Reviewer #4

(Remarks to the Author)

Overall I am satisfied by the edits and responses the authors have given to my concerns. I welcome the introduction of Figure 1 to the paper, and the treatment of gamma is now clearer.

Reviewer #5

(Remarks to the Author)

Thank you for your responses and edits. No further questions.

Version 3:

Reviewer comments:

Reviewer #6

(Remarks to the Author)

I was not involved in the initial review process and was asked specifically to assess the authors' response to a comment raised another reviewer; hereafter referred to by Reviewer 3.

The manuscript proposes a novel hybrid modelling framework for predicting kidney graft failure based on pre transplantation data and early post operative graft function, quantified via the glomerular filtration rate. The authors combine machine learning techniques with a phenomenological simulation model and validate their approach using two independent patient cohorts with no overlap. The results demonstrate accurate and robust prediction of graft failure.

Reviewer 3 criticises the approach as home made and insufficiently benchmarked against the state of the art, concluding that the manuscript should not be considered. Specifically:

The state of the art in nephrology and robust modeling longitudinal process jointly with time-to-event are still missing. The authors seemed to propose an home-made methodology. They have to demonstrate its performance against the state of the art, for instance by simulations. Such an application does not allow us to validate the proposal. I believe that this paper should not be considered.

In response, the authors provide validation of two key components of their framework. First, they compare the overall predictive performance of their model against two established state of the art classifiers and demonstrate superior performance; this was already included in the manuscript. Second, they newly added a comparison of their glomerular filtration rate estimator at day 356 post-transplantation against three linear mixed models, showing comparable accuracy.

In my view, this response adequately and convincingly addresses Reviewer 3's concern. Linear mixed models are a widely used and currently popular approach in nephrology, and the added comparison constitutes a meaningful benchmark and improves the manuscript substantially.

I would, however, like to provide additional comments on Reviewer 3's request, as I believe that, while it refers to a fundamental aspect of robust longitudinal modelling, there may be a misunderstanding between Reviewer 3 and the authors. Unfortunately, the original comment from Reviewer 3 leaves some room for interpretation. It may reasonably be inferred that the reviewer refers to recent work on mixed effects longitudinal survival models for slope based endpoints in chronic kidney disease trials, which explicitly address the challenge of informative censoring due to event related dropout [see for example [1]]. While such models are still an area of active methodological debate [see for example [2]], it is well recognised that misspecification of censoring mechanisms can induce substantial bias.

The authors clearly state that their modelling strategy deliberately avoids informative censoring by restricting analysis to patients without graft rejection. They justify this choice and support it with cross validation. Within this explicitly defined framework, a comparison with linear mixed models is appropriate and sufficient, even though it does not address informative censoring effects or "longitudinal process jointly with time-to-event" modelling approaches.

I disagree with the assertion that a custom-built methodology cannot be adequately validated. The comparisons and validation procedures for the model as a whole as predictor for graft failure provided by the authors are rigorous and proportionate to the stated aims of the study. The evaluation metrics are appropriate, and the manuscript offers clear potential value for personalised patient management in a clinical setting ultimately resulting in an potential improvement of patient care. I believe that this manuscript will be received with interest not only by nephrologists, but also by the wider transplantation community and researchers in predictive modelling.

I therefore recommend this manuscript for publication.

Addendum:

While I mainly focussed on the response of the authors, I have two minor comments the authors may want to consider:

Main manuscript, line 172: The abbreviation GFR has already been introduced in line 50.

Supplementary material, line 283: The sentence appears incomplete or lacks appropriate punctuation.

References:

[1] E. Vonesh et al., Mixed-effects models for slope-based endpoints in clinical trials of chronic kidney disease, *Statistics in Medicine*, 2019

[2] C. Thomadakis et al., Longitudinal and time-to-drop-out joint models can lead to seriously biased estimates when the drop-out mechanism is at random, *Biometrics*, 2019

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Dear Editor,

Thank you for your valuable feedback on our manuscript. We would like to express our sincere gratitude for the time and effort you invested in reviewing our manuscript, " Identification of a Personalized eGFR Threshold Improves the Prediction of Kidney Failure Risk After Transplantation." submitted to Communications Medicine Journal (COMMSMED-23-0937-T). The reviewers' constructive feedback and insightful comments have been instrumental in enhancing the quality and clarity of the manuscript. We have addressed all these comments and both a clean and marked-up copy of the manuscript with supporting information have been submitted along with this review. Please note that the reviewer's comments are given in black bold font and our replies are in blue. The structure of this report is given in the following content section

Contents

Response to Reviewer 1's Comments	2
Response to Reviewer 2's Comments	12
Response to Reviewer 3's Comments	20
Response to Reviewer 4's Comments	21
Major changes	28

Response to Reviewer 1's Comments

1.1.13. Fig. 1 – the distributions look the same or very similar, with almost entirely overlapping interquartile range. How are there significant differences?

Many thanks for your comment. The figure changed a bit, and medians with interquartile ranges are shown. Kruskal Wallis with Mann Whitney were applied to the distributions. Although the distributions of the groups are visually similar with overlapping interquartile ranges, higher-order moment analysis indicates significant differences between the groups. This suggests that while central tendency and spread are comparable, the groups exhibit variations in skewness, kurtosis, and other higher-order statistical moments, revealing deeper structural differences in the data.

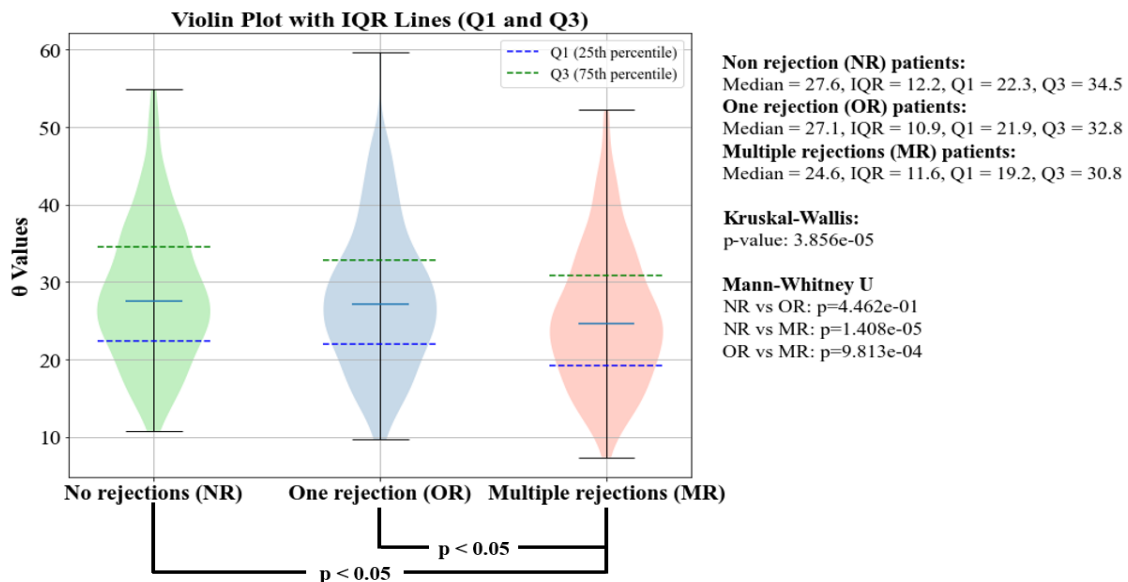


Figure 1.1. Updated Figure 1

Thus, we added the following text in section 2.4.:

“Although the distributions of the groups are visually similar with overlapping interquartile ranges, higher-order moment analysis indicates significant differences between the groups. This suggests that while central tendency and spread are comparable, the groups exhibit variations in skewness, kurtosis, and other higher-order statistical moments, revealing deeper structural differences in the data.”

1.2.13. The results in 2.4 are unclearly presented with respect to comparing the proposed method to XGBoost and Random Forest since those ML algorithm results are not broken down in the same way as the proposed methods as illustrated in Fig 3. From the description it seemed like the authors proposed method benefitted from additional tuning to the data, than would have been available for the two ML algorithms used. Furthermore, there is no mention of the authors applying a hyperparameter sweep for XGBoost or Random Forest to give these algorithms the best chance of success (standard practice for ML modeling). Presumably default run parameters were used for both ML algorithms, but additional details about how these algorithms were applied (e.g. what hyperparameter settings, and clarify any differences in how the proposed method in this paper vs. these two ML algorithms are being applied as a GF classifier).

Thank you for your thorough assessment and chance to improve. In the amended manuscript, we addressed issues about comparing the suggested technique to XGBoost and Random Forest. We optimised both algorithms' hyperparameters for a fair and accurate comparison.

The parameter grid for hyperparameter optimization in XGBoost was:

```
parameter_grid = {'n_estimators': [100, 200, 300], 'max_depth': [3, 4, 5], 'learning_rate': [0.01, 0.1, 0.2], 'subsample': [0.8, 0.9, 1.0], 'colsample_bytree': [0.8, 0.9, 1.0]}
```

Random Forest uses the following parameter grid:

```
parameter_grid = {'n_estimators': [100, 200, 300], 'max_depth': [3, 4, 5], 'min_samples_split': [2, 5, 10], 'min_samples_leaf': [1, 2, 4], 'bootstrap': [True, False]}
```

Based on the updated tables S.7-S.9. given below, we found that hyperparameter tuning improved model accuracy from 0.54 to 0.89 . The regions marked in red indicate a decrease in accuracy, while regions marked in green indicate an increase in accuracy with the parameter optimization. In particular, patients with multiple rejections had reduced accuracies. These adjustments should alleviate your concerns and make the proposed method's comparison to machine learning techniques more transparent and robust.

Table S. 7: XGBoost, and Random Forest accuracies for different critical thresholds ($Crit_{thr}$) in non-rejection patients. The F1 score is in parentheses.

Classifier	AUC with F1 score for $Crit_{thr}$ equal to 30 mL/min/1.73m ²	AUC with F1 score for $Crit_{thr}$ equal to 40 mL/min/1.73m ²	AUC with F1 score for $Crit_{thr}$ equal to 50 mL/min/1.73m ²
XGBoost	0.65 (0.59)	0.73 (0.75)	0.76 (0.76)
Random Forest	0.73 (0.66)	0.74 (0.73)	0.79 (0.79)

Table S. 8: XGBoost, and Random Forest accuracies for different critical thresholds ($Crit_{thr}$) in one rejection patients.

Classifier	AUC with F1 score for $Crit_{thr}$ equal to 30 mL/min/1.73m ²	AUC with F1 score for $Crit_{thr}$ equal to 40 mL/min/1.73m ²	AUC with F1 score for $Crit_{thr}$ equal to 50 mL/min/1.73m ²
XGBoost	0.73 (0.76)	0.74 (0.70)	0.83 (0.88)
Random Forest	0.88 (0.86)	0.68 (0.66)	0.87 (0.87)

Table S. 9: XGBoost, and Random Forest accuracies for different critical thresholds ($Crit_{thr}$) in multiple rejection patients.

Classifier	AUC with F1 score for $Crit_{thr}$ equal to 30 mL/min/1.73m ²	AUC with F1 score for $Crit_{thr}$ equal to 40 mL/min/1.73m ²	AUC with F1 score for $Crit_{thr}$ equal to 50 mL/min/1.73m ²
XGBoost	0.73 (0.74)	0.72 (0.72)	0.74 (0.75)
Random Forest	0.75 (0.74)	0.68 (0.69)	0.71 (0.73)

Additionally, we changed the latest paragraph of section 2.5 to

“Finally, by using the threshold concept, we examined if state-of-the-art classification methods such as XGBoost and Random Forest could outperform our algorithm. XGBoost was performing between 0.75 ± 0.06 (F1-score: 0.74 ± 0.07) across patient groups and for different thresholds and Random Forests between 0.75 ± 0.07 (F1-score: 0.75 ± 0.08) (for details see section 5 of Supplemental Material 2). Therefore, the coupling of regression, mathematical modeling and uncertainty analysis had a better accuracy with respect to patient classification.”

Also, we updated the section 5 of supplemental material 2 by adding the following text:

“For classification XGBoost and Random Forest were performed in the patient with one, with multiple and without rejection, and the NearMiss algorithm was used due to imbalanced data. The hyperparameter optimization for the XGBoost model was conducted using the following parameter grid: the number of estimators was set to [100, 200, 300], the maximum tree depth ranged from [3, 4, 5], the learning rate was varied across [0.01, 0.1, 0.2], the subsample values were chosen from [0.8, 0.9, 1.0], and the column sampling by tree values were selected from [0.8, 0.9, 1.0]. For the Random Forest model, the hyperparameter optimization utilized a parameter grid consisting of: the number of estimators set to [100, 200, 300], maximum depth options of [3, 4, 5], the minimum number of samples

required to split a node ranging from [2, 5, 10], the minimum number of samples per leaf set to [1, 2, 4], and both bootstrap settings [True, False] were evaluated.”

Lastly, we modified the sentence below Table S.9. by writing:

“Based on the results on Table S. 7-S. 9, regardless of considering the issue of data imbalance, on average, both the ROC AUC and F1 scores were lower than those achieved by our procedure.”

1.3.13. Applying cross validation vs. a single partitioning of the original data into a training and validation cohort, since the results are open to sample bias (i.e. the instances in the validation set could be easier or harder to predict accurately just by chance).

Many thanks for your question. Our study is structured into three main parts, as illustrated in the updated Figure 5. The first part involves multilinear regression to predict the GFR (glomerular filtration rate) after one year. The second part focuses on personalized dynamic model parameter estimation, dynamic model cannot be treated like a statistical model. The final part deals with model-based uncertainty analysis and classification. Cross-validation was applied specifically to the first part, the multilinear regression, to mitigate the risk of sample bias that can occur with a single partitioning of data. For the fitting part we use the personalized patient prediction of the regression to calibrate the dynamic model parameters.

Thus, we added the following text in section 2.1.:

“The first part involves multilinear regression to predict the GFR (glomerular filtration rate) after one year (Machine Learning). Cross-validation was applied to the multilinear regression, to mitigate the risk of sample bias that can occur with a single partitioning of data. The second part focuses on personalized dynamic model parameter estimation for each patient based on the regression results (Mathematical Modeling). The final part deals with identifying the sources and magnitude of model uncertainty analysis and the classification of patients according to their GFR threshold regarding graft survival (Monte-Carlo).”

1.4.13. Linked to the above comment: -How were the training and validation cohorts determined? Did you start with a single dataset and randomly partition the instances? Did they represent two patient groups collected over different time intervals? These details would be necessary to consider other potential limitations of the presented analysis given that only a single data partition was utilized.

Thank you for your insightful question. For our study, we utilized an extended dataset of patients from the Hannover Medical School, covering two distinct time periods. Specifically, we used a dataset of 1739 patients transplanted between the years 2000-2015 and splitted the dataset in half (Cohort 1-Training; transplanted 2000-2008, cohort 2-Validation; transplanted 2009-2015). Please note that the two cohorts involve different patients. We used this temporal division to create the training and

validation sets, ensuring that each cohort represents a distinct group of patients collected over different time intervals. This method allowed us to preserve the integrity and chronological order of the data, which is crucial for understanding the temporal dynamics and potential changes in medical practices or patient demographics over time. By splitting the data in this manner, we aimed to mitigate potential biases that could arise from random partitioning, ensuring that each subset reflects different periods and possibly different clinical conditions and treatments. This approach provides a more robust validation of our model's performance and helps identify any temporal trends or shifts in patient outcomes. However, we acknowledge that using a single partition may have its limitations, such as the potential for unobserved confounders unique to each time period. Future work could involve additional cross-validation within each cohort to further validate our findings and ensure their generalizability across different patient groups and time periods.

Thus, we added the following text in section 4.1.:

“Data was taken from the Hannover Medical School, covering two distinct time periods. Specifically, the dataset of 1739 patients transplanted between the years 2000-2015 was splitted in half (Cohort 1-Training; transplanted 2000-2008, cohort 2-Validation; transplanted 2009-2015). This temporal division was applied to ensure that each cohort represents a distinct group of patients to preserve the integrity and chronological order of the data, which is crucial for understanding the temporal dynamics and potential changes in medical practices or patient demographics over time. By splitting the data in this manner, potential biases that could arise from random partitioning were mitigated, ensuring that each subset reflects different periods and possibly different clinical conditions and treatments. This approach provides a more robust validation of our model's performance and helps identify any temporal trends or shifts in patient outcomes.”

1.5.13. Is the accuracy in this study based on a binary class outcome i.e. rejection vs. no rejection, or multiclass, i.e. no rejection, one rejection, multiple rejections? This is not always clear in the paper from section to section and figure to figure. Either way it seems like the target outcomes is imbalanced (as mentioned in the supplemental materials), meaning that standard accuracy, and AUC of ROC may not be the fairest evaluation criteria, due to the class imbalance. Authors should consider balanced accuracy and AUC-PRC as an alternative or in addition to ROC-AUC.

Thank you very much for your valuable comment. The final outcome is a binary indicator of the risk of kidney failure, determined by comparing the patient's GFR at one or two years post-transplant with a predefined GFR threshold. We incorporated the F1 score in our analysis, which enhances the evaluation of classification models, particularly in the presence of imbalanced classes.

We made the following additions to the manuscript.

1) We updated the final paragraph of section 4.5:

“The final outcome is a binary measure of the risk of kidney failure, determined by comparing the individual's status at one and two years. Status is defined as the return to dialysis once the GFR reaches a specified threshold, referred to as GFRfail, level (30-50 mL/min/1.73 m2). A comprehensive analysis

of how this threshold is estimated can be found in Section 2 of the Supplemental Material 2. The corresponding area under the curve (AUC) was calculated for each ROC in order to select the best parameter set that represents the individualized risk. Also, the F1 score was incorporated into the analysis to address the issue of class imbalance in the final outcomes.”

2) In Figures 4 and 5, we have included the F1 scores from our analysis, and their respective trends are discussed in detail in Sections 2.5 and 2.6 of the text.

3) When we applied the classical classifiers (XGBoost & RF), we used an undersampling technique to deal with the class imbalance problem.

1.6.13. From section 4.3, it seems possible that interpolation of GFR values could impact the presented results. Can the authors include discussion on this consideration? Are there any strategies that could be employed to account for this possibility (or avoid them, using the data as is without interpolation?)

Thanks for your thoughtful question. Indeed, GFR interpolation may affect results, which merits further explanation. When measurements are far apart, interpolation might induce biases and errors. One solution is to boost GFR measurements, especially in the first 100 days post-transplantation. By collecting more data points at this important period, interpolation would be decreased, improving model predictions. Our approach's reliance on interpolated GFR data is a limitation. Interpreting our findings requires considering interpolation's impact. This constraint emphasizes the need for a large dataset with regular and consistent measurements. Our outcomes' susceptibility to alternative interpolation methods should be examined in future research. Comparing results using different interpolation approaches might help us assess our findings' robustness and locate the best missing data handling solutions. Linear interpolation, spline interpolation, and advanced machine learning methods may be used in this investigation. To reduce interpolation, we need to boost clinical measures. Another way to improve the situation is by using Gaussian Processes that assess the uncertainty of the interpolated points. This method improves GFR forecasts and gives a more complete picture of the patient's health. In conclusion, interpolating GFR data is important in our methodology, but we realize its limitations and potential impact on our results. To improve model robustness and accuracy, future research should measure more often and include more clinical data. We must also examine the sensitivity of outcomes to different interpolation methods to improve our approach and predictability.

Thus, we added the following part in the limitations section in the fifth paragraph of section 3:

“Furthermore, our outcomes' susceptibility to alternative interpolation methods should be examined in future research. Comparing results using different interpolation approaches could help to assess our findings' robustness and locate the best missing data handling solutions. Linear interpolation, spline interpolation, and advanced machine learning methods may be used in this context. To reduce interpolation, boost clinical measurements should be frequent and consistent. Another way to improve the situation is by using Gaussian Processes that assess the uncertainty of the interpolated points.”

1.7.13. It's not clear how generalizable this solution would be. Would this method need to be adapted to different patient populations, e.g. when applied at different clinical institutions. Do the authors expect or have plans to evaluate this model on other patient samples, and larger sample sizes? If so would this algorithm need to be retrained or reoptimized for different populations?

Thank you very much for your insightful comment. Initially, our analysis was conducted using two large cohorts. The two cohorts belong to different time eras (200-2008 and 2009-2015) and this could account for robustness in the different treatment protocols. To assess the generalizability of our findings, we performed several sensitivity analyses across both cohorts, covering one- and two-year periods. These analyses involved stratifying the populations by patients' sex, donors' sex, and patient age. The results, measured through the AUC and F1 score metrics, demonstrated consistent performance across these subgroups, supporting the robustness of our model. However, as part of future work, this analysis should be extended to additional cohorts that include patients from different hospitals (undergoing different treatment protocols), individuals from diverse racial backgrounds, and those with severe comorbidities. Such expansion would allow us to further evaluate the generalizability of our findings and enhance the overall robustness of the analysis.

Thus, we added the following part as a last paragraph of section 2.6 in the main manuscript:

"In order to evaluate the generalizability of our approach, we conducted a sensitivity analysis by stratifying both the training and validation cohorts based on patient sex, donor sex, and patient age. Subsequently, we performed our analysis over one- and two-year periods for each of these subgroups. The results, as measured by the AUC and F1 scores, are presented in Table 3 and Table 4 for the training and validation cohorts, respectively. Both the AUC and F1 score values demonstrated consistent performance across the different strata, underscoring the robustness and reliability of our methodology. This sensitivity analysis further highlights the strength of our approach in diverse patient populations and timeframes."

Also, we added in the last paragraph of the discussion section 3, the following text:

"Such analysis could be extended to additional cohorts that include patients from different institutions (undergoing different treatment protocols), individuals from diverse racial backgrounds, and those with severe comorbidities. Such expansion would allow us to further evaluate the generalizability of our findings and enhance the overall robustness of the analysis."

1.8.13. Line 99 – reference error.

Many thanks for this comment. We removed the reference error.

1.9. Supplemental materials are somewhat difficult to follow in the context of the provided main paper text and additional supplemental materials text. I'd suggest providing more context and explanation in the supplemental materials if there is no space in the main text.

Many thanks for your thoughtful feedback on the supplemental materials. We understand the importance of clarity and cohesion between the main manuscript and the supplementary sections, and we greatly appreciate your suggestion to enhance the contextual explanations. In response, we have made several improvements to create a stronger connection between the main manuscript and the supplemental materials. Specifically, we added additional text and clarifying explanations within the supplementary documents to provide a more comprehensive background and a clearer link to the main paper's content. We've included both the revised manuscript and supplemental documents to highlight the changes made, which we believe address the clarity concerns effectively. We hope these enhancements will offer a more cohesive and accessible experience for readers. Thank you again for your valuable input.

1.10.13. Fig 2C, not enough context is provided in the paper to explain what this figure depicts clearly. I'd check all figure and table legends and update them as needed to provide greater context as needed.

Many thanks for your comments. We appreciate your feedback. We have added additional text to Figure 3C and updated the context to provide more information for the readers. We have also reviewed and updated all figures and table legends as needed to ensure they provide sufficient context.

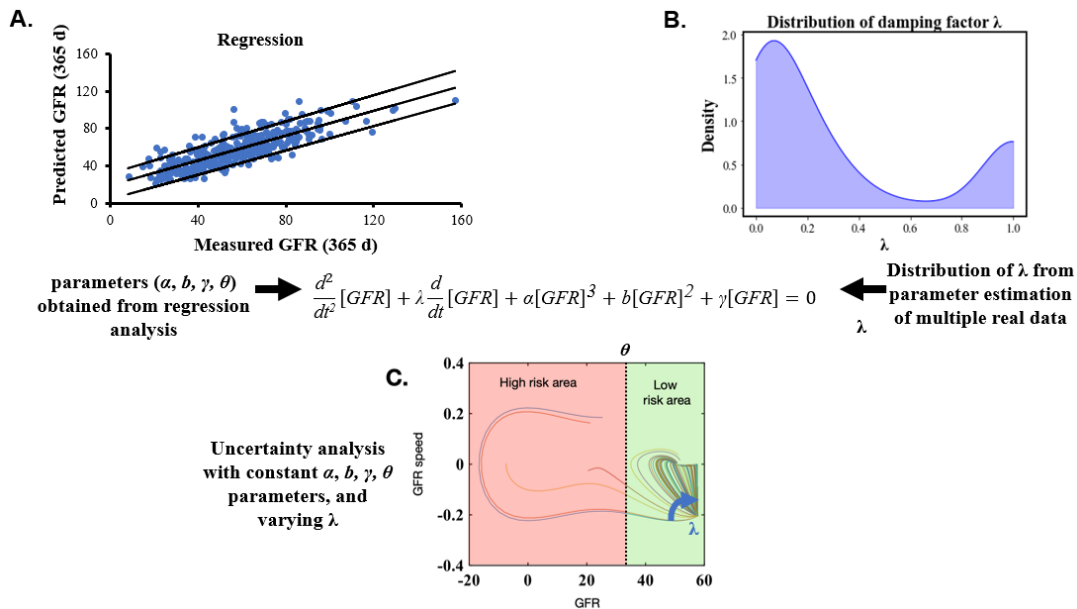


Figure 1.2. Updated figure 2C

More precisely we mentioned in section 2.5:

“Our regression analysis, integrated with the ± 20 -unit uncertainty in critical GFR thresholds (θ) detailed in Section 2 of Supplementary Material 2, consistently applicable across all patient categories. Utilizing this prediction interval, the minimum (GFR_{min}) and maximum (GFR_{max}) GFR values were applied to derive two additional parameter sets for α , β , and θ in accordance with the Eqns. 10-12 (Figure 3A). Regarding the damping factor λ , it remains subject to uncertainty. As stated in the section 4.3. of Material and Methods, the damping parameter λ was fitted from the individual GFR measurements of non-rejection patient ensemble, including derivation of the corresponding probability distribution (Figure 3B). Since only the empirical distribution of the parameter λ was known, we generated an ensemble of simulations for λ values drawn from this distribution, and the probability of graft failure was calculated (Figure 3C). In this figure, the probability of graft failure corresponds to the GFR transitions in the pink high-risk area, where the GFR falls below the critical threshold (θ), existing at the interface between the pink high-risk area and the green low-risk area. This analysis was applied three times, first for the predicted GFR from the regression, then for the minimum (GFR_{min}) and maximum (GFR_{max}) GFR values, across two populations (N1 and N2), predicting graft risk at 365 days and 730 days post-transplantation.”

1.11.13. I found it a little confusing to re-name the training and validation datasets as N1 and N2 groups. Linked to this, on line 343, the authors mention they validate their predictions in two independent datasets, but only one validation dataset was apparently described and utilized.

Many thanks for your comment. We appreciate your attention to detail. You are correct that further clarification is needed regarding our methodology on line 343. To clarify, we validated our uncertainty analysis (Monte Carlo) using both the training (N1) and validation (N2) datasets. The first cohort includes patients whose first transplantation occurred between 2000 and 2008, while the second cohort comprises patients whose first transplantation took place between 2008 and 2016. We employed this temporal division to create the training and validation sets, ensuring that each cohort represents a distinct group of patients collected over different time intervals. This approach helps to mitigate sample bias and provides a more robust evaluation of our model's performance by reflecting the temporal variations and potential changes in clinical practices over time.

Therefore, we added to the text in the section 2.1.:

“The most important model parameter is the GFR threshold value. GFR values above this threshold indicate a stable, functioning graft, whereas values below suggest a high risk of graft failure. Two cohorts were used in our analysis: a training cohort (N_1) and a validation cohort (N_2). The first cohort includes patients transplanted between 2000 and 2008, while the second cohort comprises patients transplanted between 2009 and 2015. We employed this temporal distinction to create the training and validation sets, ensuring that each cohort represents a distinct group of patients collected over different time intervals. This approach helps to mitigate sample bias and provides a more robust evaluation of the model's performance by reflecting the temporal variations and potential changes in clinical practices over time. The model was calibrated exclusively using patients without rejections in the training cohort, as their measured GFR values were less influenced by the use of immunosuppressant drugs, which are commonly administered in the treatment of rejections. This decision also helped us avoid introducing bias or data leakage into the analysis. Using uncertainty analysis techniques, we

calculated the probability of graft failure in the two independent datasets, which included patients with no, one, or multiple rejections, using a small number of input variables. Our three-step strategy is outlined in Figure 1.”

1.12.13. Section 3 mentions the use of state of the art feature selection techniques. What makes the ones used, ‘state-of the art’, since they seemed fairly traditional based on my experience.

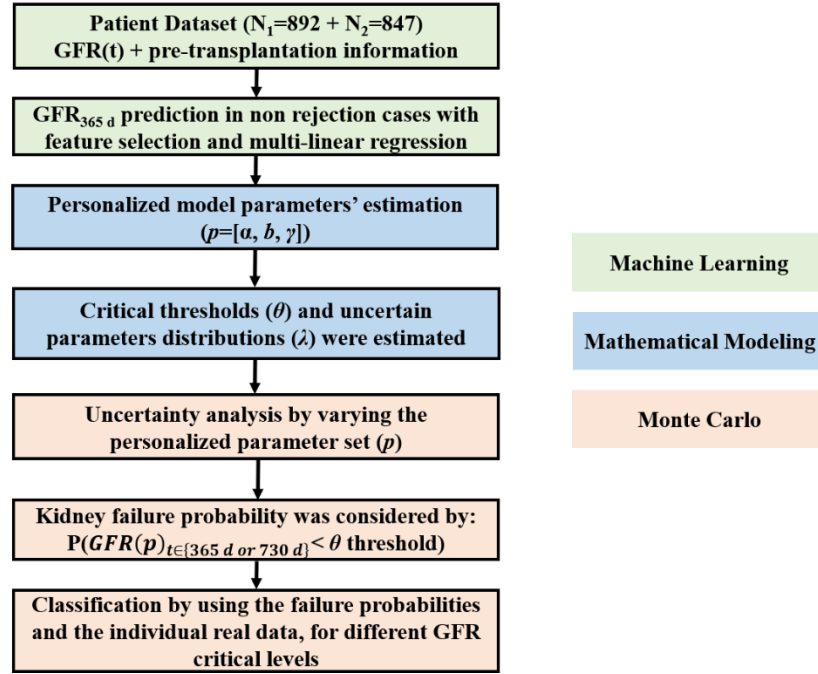
We appreciate your feedback. Actually, our feature selection methods are typical after assessment. Thus, we omitted "state-of-the-art" from our description in the first paragraph of the section 3. The features in our multilinear regression model contain essential biological inputs linked to graft function. These biological inputs were chosen for their known relevance for graft health and performance. By adding biologically significant characteristics, we hope to improve the model's prediction power and graft function assessments over time.

Thus we modified the following sentence in the first paragraph of section 4.2:

“The multilinear regression model included pre-transplantation biologically relevant data and modellable parameters, including”

1.13.13. Figure 5 is a bit distracting with the black background and white text. This figure could be more clearly constructed overall.

Many thanks for your comment. We changed the font from black to white a with black bold letters. Also, we showed the three main stages followed (machine learning, mathematical modeling and uncertainty analysis).



Updated Figure 1. Steps followed from experimental measurements to the classification of the patients. Section 4.3 includes model details regarding the model parameters α , b , γ , θ , and λ .

We have also revised the text in response to the comments from Reviewer #4 and added the following to the last paragraph of Section 2.1:

“Our three-step strategy is outlined in Figure 1. The first part involves multilinear regression to predict the GFR (glomerular filtration rate) after one year (Machine Learning). Cross-validation was applied to the multilinear regression, to mitigate the risk of sample bias that can occur with a single partitioning of data. The second part focuses on personalized dynamic model parameter estimation for each patient based on the regression results (Mathematical Modeling). The final part deals with identifying the sources and magnitude of model uncertainty analysis and the classification of patients according to their GFR threshold regarding graft survival (Monte-Carlo).”.

Response to Reviewer 2's Comments

2.1.13. First of all, their goal is to use 2 early GFR measurements in KT recipient (KTRs) to predict the function of the graft at 1 and 2 years. In this era, the short term survival of allografts is believed to be excellent, with 1 year survival of 98% and 3 years of 95%. Allograft dysfunction and failure often happen later on (ie in the 2nd decade of the allograft life), and that's where we need to improve outcomes. Year 1 and 2 allograft function - are not necessarily indicative of what

happens long term with longer exposure to CNIs, later rejections, weight gain, worse DM control, etc.

Many thanks for your comment. No doubt that in the long-term there are additional and likewise important factors that affect the graft outcome. On the other hand, early graft dysfunction or low function affects the long-term outcome and therefore should be considered as important.

We added these sentences in the fifth paragraph discussion section 3 of our article:

“Finally, our methodology cannot foresee complications that independently occur in the long-term after transplantation. However, GFR from the first transplant year is highly predictive of the future course of graft function and graft survival (First MR, 2003, DOI: 10.1093/ndt/gfg1027).”

2.2.13. Moreover, this study spans over 15 years for KTRs, in which kidney allograft survival has significantly improved with changes to the medications, surgical techniques, and so function from the authors testing or validation cohorts for early 2000s does not apply now. The authors do mention the change from CYA to Tac during their study period, but no details are given to how many pts were in each group.

Many thanks for your insightful comment. We have seen some gradual improvement in graft survival during the last two decades. We would rather see the heterogeneity in treatment as an advantage, since our model apparently is *robust and applicable to different treatment era's*. We provided more details regarding the proportion of patients on different immunosuppressants.

Thus, we added the following text, at the end of the second paragraph of subsection 4.1.:

“A detailed description of the pre- and early post-transplant variables of the cohorts, including the initial treatment with immunosuppressants, has been previously reported in the work of Scheffner et al. (DOI: 10.1097/TP.0000000000002922).”

As for older therapies/procedures, we added in the first paragraph of the discussion section 3:

“Our framework, spanning patients from two previous decades, is apparently independent of gradual changes in procedures and medical treatments, as demonstrated by the validation results of the model.”

2.3.13. There is also allot of clinical data missing, such as levels for CNIs in the timeframe of the GFR tests, which are usually intentionally higher in the first 3 months (where their first and sometime 2nd calculations is from), and can cause the GFR to go down.

Many thanks for your insightful comment. The focus of our work was to define critical eGFR values and eGFR evolution. It was not the aim to build a complex multifactorial model with serial data on immunosuppressant's blood levels.

2.4.13. Additionally, their groups are divided based on rejection yes/no/multiple in the first 6 months after the txp (I am assuming, as the time for the rr), but we know that earlier rejections are often treatable and are not necessarily associated with later function. Also - while the authors so mention types of rejection and using the Banff score for grading, there is no data of type and grade of rejections in the cohort (specifically rate of each type), which makes it harder to extrapolate - treated early TCMR Banff 1a outcomes are not the same as TCMR grade 2 or ABMR.

Thank you for your comment. For the purposes of this study, we treated both biopsy types equivalently to maintain consistency and simplify interpretation without introducing additional variables that would complicate the statistical analysis.

The following Table 1 is added in the supplementary material 1, and shows the number of biopsies with acute TCMR and ABMR within the first year post-transplant (Tx) in both the training and validation groups, categorizing the rejections into BL-TCMR, TCMR IA/B, and TCMR IIA/B and ABMR.

Table 1. Number and percentage (in parentheses) of biopsies with acute TCMR and ABMR within the first year after transplantation in the training and validation groups.

	Training group N=3147 biopsies	Validation group N=2495 biopsies
Acute TCMR, n (%)		
BL-TCMR	466 (14.8)	431 (17.3)
TCMR IA/B	294 (9.3)	101 (4.0)
TCMR IIA/B	114 (3.6)	87 (3.5)
Acute ABMR, n (%)	398 (9.0)	294 (11.7)

We added the following text in a second paragraph of section 2.2.: “In supplementary material 1, Table 1 details the number of biopsies with acute TCMR and ABMR within the first year post-transplant in both the training and validation cohorts. Acute TCMR is subdivided into BL-TCMR, TCMR IA/B, and TCMR IIA/B.

2.5.13. Line 352+353 notes that these cohorts included KT alone and KTs combined with other SOT in the methods section. These groups do not behave the same, and there is only one mention later on of KT vs SPK, which also differ as at least in the US - SPK organs are often of better quality. But, I am also left with the question of whether or not there were other SOT but pancreas included here, and how many?

Many thanks for your insightful question. The data related to the inclusion of patients with simultaneous organ transplantation (SOT) are now provided in the Supplemental Tables. We fully agree that patients with combined SOT, such as simultaneous pancreas-kidney (SPK) transplants, may exhibit different outcomes compared to those receiving kidney transplants (KT) alone. We acknowledge that the

inclusion of these patients contributes to an increase in cohort heterogeneity, which we view as an advantage for enhancing the generalizability of our findings. Furthermore, while pancreas transplant recipients were the most frequent combined SOT group, we have also included data on liver, heart, and lung transplantations, as shown in the Tables 2 and 3 of supplementary material 1.

Table 2: Transplantations per different groups in the training dataset

	No rejection	One rejection	Multiple rejections	p-values
N	354	260	278	
Pancreas transplantation				
Yes	28 (7.9)	19 (7.3)	18 (6.5)	0.79
No	326 (92.1)	241 (92.7)	260 (41)	
Liver transplantation				
Yes	2 (0.6)	2 (0.8)	0 (0)	0.38
No	352 (99.4)	258 (99.2)	278 (100)	
Heart transplantation				
Yes	5 (1.4)	1 (0.4)	1 (0.4)	0.23
No	349 (98.6)	259 (99.6)	277 (99.6)	
Lung transplantation				
Yes	2 (0.6)	2 (0.8)	1 (0.4)	0.81
No	352 (99.4)	258 (99.2)	277 (99.6)	
Multiorgan transplantation				
Yes	36 (10.2)	24 (9.2)	20 (7.2)	0.42
No	318 (89.8)	236 (90.8)	258 (92.8)	

Table 3: Transplantations per different groups in the validation dataset

		No rejection	One rejection	Multiple rejections	p-values
N		410	256	181	
Pancreas transplantation					
	Yes	36 (8.8)	10 (3.9)	3 (1.7)	0.00
	No	374 (91.2)	246 (96.1)	178 (98.3)	
Liver transplantation					
	Yes	2 (0.5)	3 (1.2)	0 (0)	0.27
	No	408 (99.5)	253 (98.8)	181 (100)	
Heart transplantation					
	Yes	2 (0.5)	0 (0)	0 (0)	0.34
	No	408 (99.5)	256 (100)	181 (100)	
Lung transplantation					
	Yes	0 (0)	0 (0)	0 (0)	1.00
	No	410 (100)	256 (100)	181 (100)	
Multiorgan transplantation					
	Yes	40 (9.8)	13 (5.1)	3 (1.7)	0.00
	No	370 (90.2)	243 (94.9)	178 (98.3)	

We added the following text in the second paragraph of section 2.2.: “*Additionally, Tables 2 and 3 of supplementary material 2 compares transplant patients in the training and validation datasets by transplantation type (pancreas, liver, heart, lung, and multiorgan) and number of rejection episodes (none, one, or multiple).*”

2.6.13. The initially refer to a training group (N1) and validation group (N2) as having similar number of patient (around 850 per group), but only later do they note that they included only the pts with NO rejection (N=372, hence less than half the group) in the training cohort for the model.

Many thanks for your comment. We used patients without any rejection (N=354) in the regression analysis because their eGFR was less influenced by immunosuppressant drugs, which were administered to patients with rejections. Additionally, we wanted to avoid introducing bias that could arise from including patients with rejections, whose eGFR levels are affected by multiple factors. By comparing the regression of eGFR predictions using only the non-rejection patients versus all patients with 10-fold cross-validation, we found no significant improvement in the model's performance. This is likely because, in patients with rejections, multiple unknown factors influence GFR values, which a linear regression model cannot adequately capture. Results are shown in Figs. 2.1 and 2.2. The R^2 is lower when all patients are included compared to when only non-rejection patients are analyzed, as shown in the figures.

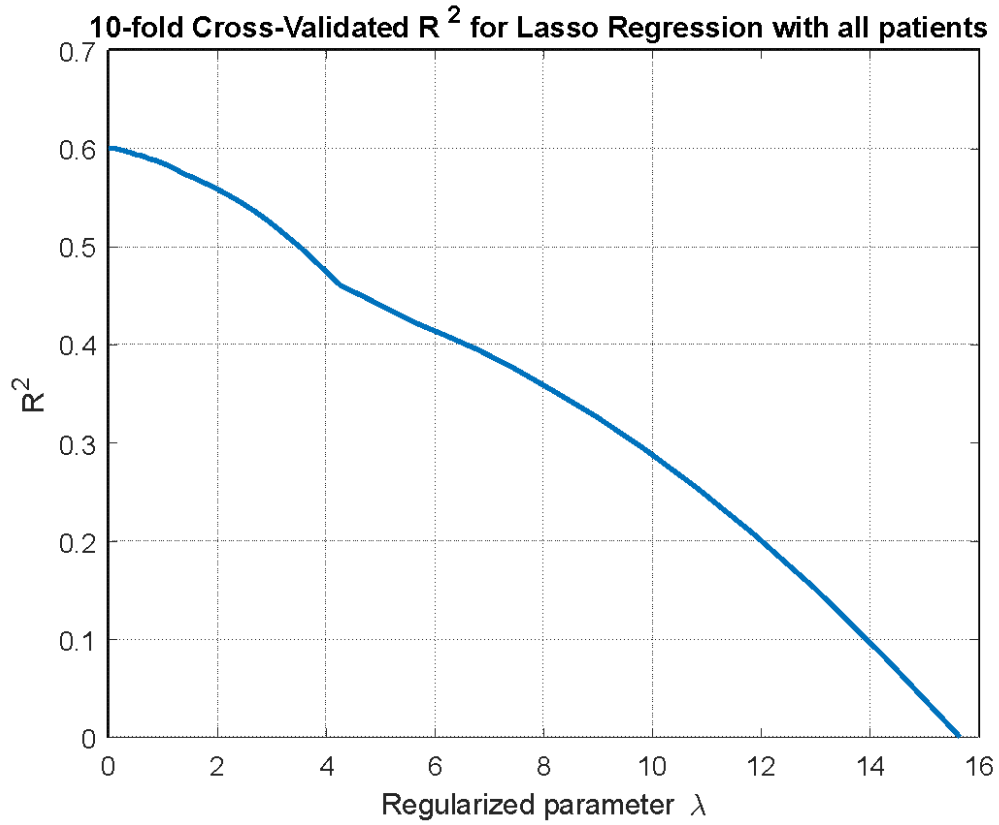


Figure 2.1. 10-fold cross validation Lasso regression for R^2 estimation, when all patients in the training group were used.

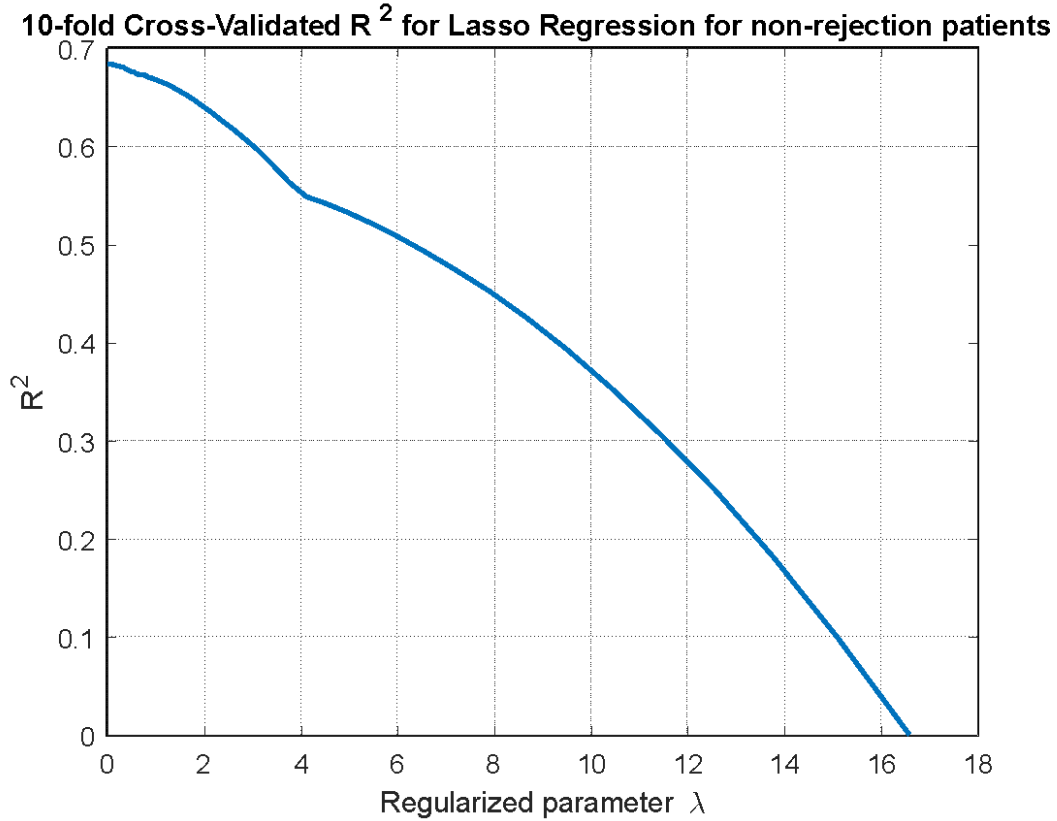


Figure 2.2. 10-fold cross validation Lasso regression for R^2 estimation, when only patients without a rejection in the training group were used.

This is the reason why we decided to work with patients without rejection only in regression analysis. This is mentioned in the last paragraph of section 2.2.:

“The inputs for the regression analysis were limited to patients without any rejection within the first year ($N=354$), as patients with rejection were assumed to have additional bias of eGFR (e.g., by severity and type of rejection, variable response to treatment of rejection), which can be hardly captured by a linear regression model. Also, we wanted to avoid any data leakage. In the feature selection, the inputs included patient age, sex, donor's age, deceased donor vs. living donor, cold ischemia time, initial function vs. delayed function of the graft, HLA-mismatch, panel reactive antibodies, blood transfusions, number of pregnancies, number of previous transplantations, highest GFR within six weeks, and percentage change between the highest GFR within six weeks and the GFR at 100 days (Section 1 in the supplemental material 2)”

2.7.13. Missing data of BK - specifically if BK was seen on bx (BK nephropathy), as a cause of GFR decline.

We appreciate your comment. This is a point we already discussed earlier. There are different factors that can affect the graft function. But, on the other hand, we intended to use a data-driven model that is based on few GFR measurements, its absolute values and its fluctuation and critical thresholds.

2.8.13. Lines 333-334 their assumption of "undisturbed transplant course"= patients with no rejection is simply wrong. There many factors besides rejection that are not addressed here and impact the graft function, such as surgical complications, AKI, hypotension, CNI toxicity, infections (UTI, BK, and more). Not using them in the model, weakens it's strength.

Many thanks for your comment. We specified this as "*transplant course not affected by rejection*". Again, the intention was not to use different additional variables such as BK, AKI,.. to predict the GFR or graft failure (see my comment to #2.3.13).

2.9.13. Patients in the validation group had notably shorter CIT than the training group. CIT is known to be a factor of allograft outcomes.

Many thanks for the comment. The significant difference in CIT is due to the much greater proportion of living-donor transplantations in the validation cohort (see Table 1 vs. Table 2). We agree that CIT has an important bearing on the outcome. However, we consider this heterogeneity between the cohorts as an advantage in terms of generalizability of our findings.

2.10.10. There is no data re race and ethnicity in the study, and we know that race (of donor and recipient) impacts allograft function in KT.

Many thanks for your comment. We added the following information in the first paragraph of section 2.2..

"There were >99% Caucasians in each group."

2.11.13. There are several statements in the beginning of the manuscript - line 47-48 and 48-49, for example, that are too definitive and not necessarily true based on available data. An example is "In particular, the GFR is determined by the quality of transplanted organ", were in fact the quality of the organ is an important but not the sole determinant of graft function.

We agree that the GFR is determined by a multitude of factors. However, in early transplant period, donor characteristics particularly donor age prevail as reported by us previously (Kettler et al, Transplant International 2019; 32: 1341–1342) and is also confirmed by the KDPI index.

Thus, we added in the second paragraph of subsection 2.2., the following text:

“In the early transplant period, donor characteristics, particularly donor age, is the predominant factor, as previously reported by our group (Kettler et al., Transplant International 2019; 32: 1341–1342) and further corroborated by the Kidney Donor Profile Index (KDPI) index.”

2.12.13. I would use the terms chronic kidney disease and kidney failure as opposed to "chronic renal disease" and "renal failure" to make the paper more uniformed (use of kidney in the context of kidney transplant is used throughout)

Many thanks for your comment. We replaced the word renal with kidney throughout the text.

2.13.13. line 256 I believe instead of "donation" it should be "transplantation" - as this paper is about the recipients.

Many thanks for your comment. “Donation” word is replaced by “transplantation”.

Response to Reviewer 3's Comments

3.1.2. The state of the art is missing. There is number of predictive tools in kidney transplantation aiming to predict the graft function and untimely the graft failure. The author must consider this literature in the introduction, better explain the originality of their proposal, and finally report the increase of prognostic capacities related to their approach.

Many thanks for your comment. We aimed to assess the performance of these models, which are based on conventional clinical factors, without incorporating any dynamic modeling. However, a comprehensive review and comparison of all these models falls beyond the scope of this paper. Our methodology focuses on the development of a dynamic mathematical modeling tool derived from data-driven machine learning analysis. Naturally, future research and further testing on additional validation cohorts will be necessary to determine the potential superiority of our approach.

Thus we added the following paragraph in the mid of the introduction (section 1) which states several works focused on predicting GFR:

“Studies using large datasets and numerous analytical methodologies show that predictive modeling for kidney function has advanced in recent years. One important effort was Goldfarb-Rumyantzev et al.'s logistic regression and tree-based 3-year graft survival prediction system (<https://doi.org/10.1046/j.0902-0063.2003.00051.x>). Their study examined pre-transplant characteristics from over 37407 end-stage kidney disease patients receiving cadaveric kidney or kidney-pancreas transplantation. Both models, especially the tree-based approach, showed a linear correlation with graft survival, delivering a 76% positive predictive value. This study shows that pre-transplant variables can improve post-transplant predictions by building solid models for long-term

kidney allograft survival prediction. Akl et al. (10.1097/TP.0b013e31818b221f) and Rao et al. (10.1097/TP.0b013e3181ac620b) underline the potential of integrating machine learning and comprehensive risk indices to estimate graft survival and failure risk. These papers show how predictive analytics in kidney transplantation is improving, enabling more tailored and successful treatment approaches. Greco et al. (<https://doi.org/10.1016/j.transproceed.2010.03.061>) employed data mining algorithms to develop predictive models for graft failure in kidney transplant patients, analyzing interactions between acute rejection (ARE), delayed graft function (DGF), chronic allograft changes (CAN) and body mass index (BMI). The study identified patients at higher risk of graft loss, particularly those with a BMI > 24.8 kg/m² and CAN, and proposed decision trees as an effective alternative to traditional statistical methods. All these studies were based on conventional clinical factors, without incorporating any dynamic modeling.

3.2.2. In a methodological point of view, the understanding of the modelling is complex. It appears as a home-made methodology which was not validate elsewhere. I did not identify how the authors properly deal with censoring, repeated measurements, and joint modelling of the time-to-event.

We would like respectfully to emphasize that our approach is exactly the novelty of our paper. Actually, we think the uniqueness of our computational approach is the main strength of our paper. We present a framework that combines dynamics modeling and machine learning. The former takes into account the temporal dynamics of GFR and it is completely different from classical machine learning approaches in the field.

Response to Reviewer 4's Comments

4.1.10. Lines 113-120 – you performed multi-linear regression with the non-rejection patients as your training set, and your results show that the R^2 for the rejection patients is around ~0.5. Why not include regressions for the rejection patients as your training set? It seems that your analysis will thus be mostly valid for the non-rejection patients, as $R^2 \sim 0.5$ is not very high. If you include the rejection patients in your regression, perhaps there will be more than 3 major features as you found and the regression will better capture all 3 categories of patients. In short, isn't clear to me your choice of regressing only on non-rejection patients. Same for lines 406-408 – you show that the assumption isn't very good.

Many thanks for your comment. We used patients without any rejection (N=354) in the regression analysis because their eGFR was less biased by immunosuppressant drugs, which were used in patients with rejections. Also, we wanted to avoid any data leakage. By comparing the regression of eGFR predictions using only the non-rejection patients versus all patients with 10-fold cross-validation, we found that there was no significant improvement in the model's performance. This occurs because, in patients with rejections, multiple unknown additional factors influence the GFR value, which a linear

regression model cannot adequately capture. Results are shown Figs. 4.1. and 4.2.. R^2 is lower when all patients were used instead of the R^2 of non-rejection patients in the following figures. Next, uncertainty analysis was necessary in order to analyse all patients types (patients with rejections or not).

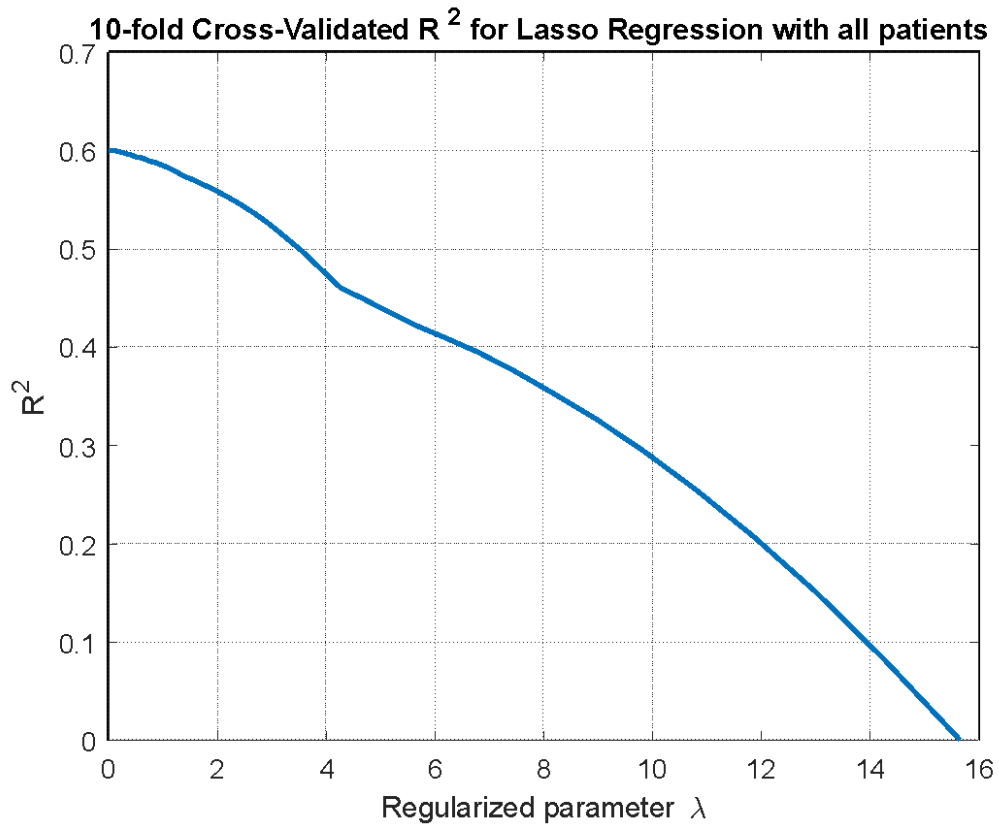


Figure 4.1. 10-fold cross validation Lasso regression for R^2 estimation, when all patients in the training group were used.

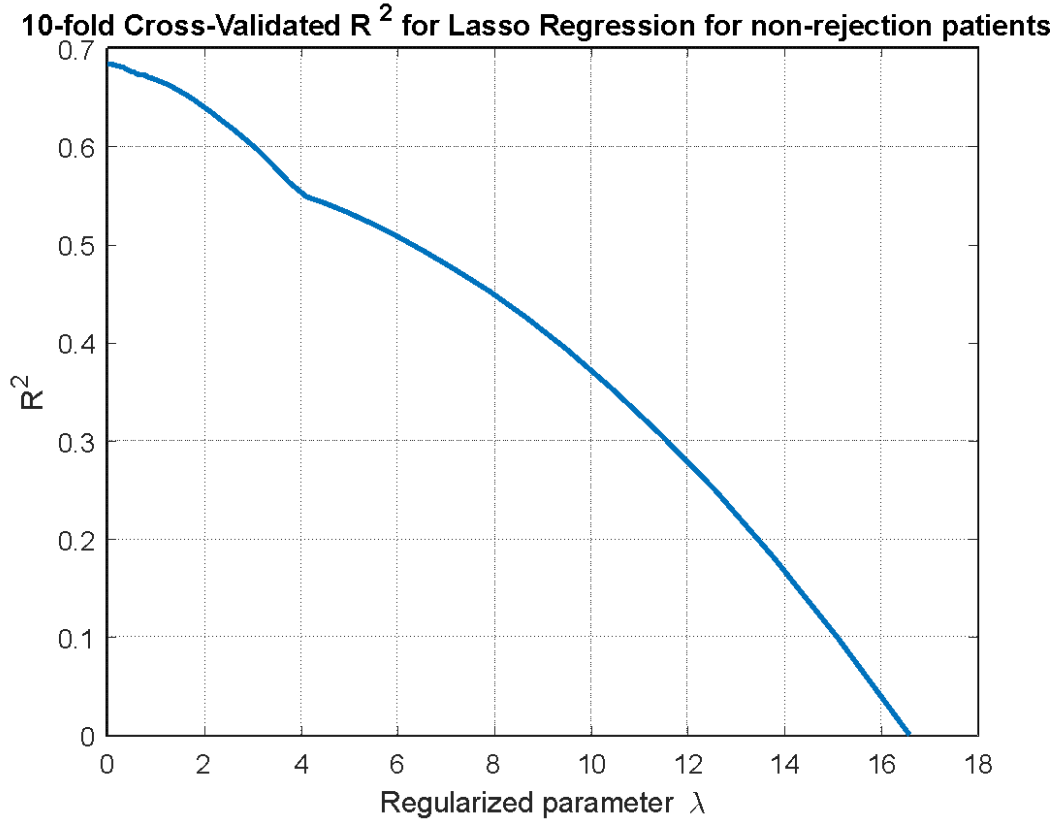


Figure 4.2. 10-fold cross validation Lasso regression for R^2 estimation, when only patients without a rejection in the training group were used.

This is the reason why we decided to work with patients without rejection only in regression analysis. This is mentioned in the last paragraph of 2.2.:

“The inputs for the regression analysis were limited to patients without any rejection within the first year ($N=354$), as patients with rejection were assumed to have additional bias of eGFR (e.g., by severity and type of rejection, variable response to treatment of rejection), which can be hardly captured by a linear regression model. Also, we wanted to avoid any data leakage. In the feature selection, the inputs included patient age, sex, donor's age, deceased donor vs. living donor, cold ischemia time, initial function vs. delayed function of the graft, HLA-mismatch, panel reactive antibodies, blood transfusions, number of pregnancies, number of previous transplantations, highest GFR within six weeks, and percentage change between the highest GFR within six weeks and the GFR at 100 days (Section 1 in the supplemental material 2)”

As for the lines 406-408, although the regression analysis did not perform optimally in patients with single and multiple rejection episodes, as indicated by a lower R^2 value and higher RMSE, the

regression output was still of significant value. It was utilized in an uncertainty analysis, where its variation contributed to improving the identification of kidney transplantation outcomes. This approach allowed us to account for the inherent variability in the data, enhancing the robustness of our predictions.

Therefore, in the same lines, we added to the text, the following:

“Although the regression model may exhibit some limitations in patients experiencing rejection episodes, potentially due to the influence of various partially unknown factors, performing an uncertainty analysis on the predicted GFR values would be essential for accurately identifying the outcome of kidney transplantation. This approach would help account for the inherent variability and provide a more reliable assessment of transplantation success. Further details on the steps involved in the uncertainty analysis are provided in Section 4.4.”

4.1a.10. Lines 37-39 in supp. material – you should reverse the order in which you state the features. According to Table S1, the most important feature is GFR_in, while in the text it seems like the donor’s age is most important.

Thank you very much for your comment. We have corrected the order of features in both the text and the Tables S.1. and S.2..

4.1b.10. Lines 59-60 in supp. Material – R^2 decreases and MSE increases. You write that both decrease.

Many thanks for your comment. We corrected it.

4.2.10. Lines 174-179 – A quantitative test to check the goodness of fit of the sigmoidal regression is missing. Furthermore, how were the survival probabilities from the Scheffner et al paper calculated? It isn’t clear what the y axis actually measured in Figures S5-8. Is survival probability itself a result of regression, or is it measurable? It is strange to see %s of survival at 15 years following transplantation without an explanation of how this survival was estimated. Were there additional inputs in the sigmoidal regression in addition to GFR365d in the original Scheffner analysis? What would happen if you included them in your regression?

Thank you very much for your comment. We did not include a discussion on the goodness of fit here because, despite the data following a sigmoid trend, the R-squared value remains quite low due to the spread of the data. To account for this uncertainty, we applied three different thresholds for GFR. However, the existence of the sigmoidal curve can be very well justified even theoretically, since this

is a typical survival cumulative distribution function. Regarding the survival probability on the y-axis, it represents the 15-year graft survival probability, which was estimated using a Random Survival Forest model based on the inputs as outlined in the work of Gwinner et al. which is still under review. More details about the work are included in the first paragraph of Section 2.3 and in Section 2 of Supplementary Material 2, with the following lines added:

“Here, we employ a survival random forest analysis developed by Gwinner et al. to assess 15-year graft survival outcomes¹⁶. By leveraging an extensive dataset from a single center¹⁷, observed over up to 15 years and including several patients, the study developed a machine-learning model to predict clinically meaningful graft outcomes. The model integrated diverse clinical and laboratory variables from the first transplant year and identified key risk factors for premature graft failure. In Gwinner et al.'s analysis¹⁶, the model demonstrated robust predictive performance, with a 71% death-censored graft survival at 15 years in the training cohort, significant concordance indices in both training (0.79) and validation (0.74) cohorts, and strong calibration. The findings underscored the model's ability to stratify patients into different risk levels for targeted monitoring and intervention strategies, emphasizing the prognostic importance of eGFR at one year, patient demographics, types of rejections, infections, and glomerulonephritis.”

4.3.10. 4.1 Approach Rationale – I strongly suggest this should feature in the introduction as it explains the approach of the whole paper. It should be among the first things someone reads when opening the paper.

Thank you for your comment. In the final paragraph of our introduction, we provided an outline of the article's structure, including a brief mention of our approach rationale, which is expanded upon in the initial paragraph of the results section (section 2). We believe this structure enhances the reader's engagement with the rationale behind our approach.

4.4.10. General comment – in the mathematical modeling section, the authors sometimes use “a” and sometimes “” interchangeably. See for example S.2 vs S.8. This happens often, should be consistent.

Many thanks for your comment. We have introduced a “.” between "S" and the number to direct the reader to a specific table or figure, equation, and table in the supplementary material.

4.5.10. Lines 120-121 in supp. Material – why do you say that alpha and gamma are small? In your derivation you didn't supply numerical values till now. Also, small compared to what? They have different units, so explain what exactly is small and why. You can also solve analytically for gamma from S.13, define $\gamma = t^2$ and solve for t. Then take square.

Thank you very much for your comment. We have conducted more thorough calculations and have derived the exact expressions that relate γ and α , as well as the threshold θ , to the estimated GFR in one year. Next, we follow certain steps to estimate γ more precisely, independently of the α , b , and θ parameters.

4.6.10. Lines 146-148 in supp. Material – in this section you take many approximations/assumptions that aren't explained.

After the insightful comments of the reviewer, we have revised completely the analysis and the parameter estimation of the model. In particular, we have recalculated all the relevant parameters (assuming steady state) without any assumptions. In turn, to estimate the GFR decay rate, we have used dimensional analysis to identify the characteristic time scale of the model which is $\sqrt{\gamma}$. The latter coincides with the GFR decay rate and then we could estimate the parameter γ from the data. Therefore, we have almost rewritten the SI part related to the above. In detail:

4.6a.10. Why do you assume that $du/dt = 0$? How is this justified.

Thank you very much for your comment. We do not use this assumption any more. The whole analysis changed substantially, and we employed an alternative methodology for estimating the γ parameter, as described in Section 2 of Supplementary Material 2.

4.6b.10. Assuming $du/dt = 0$, you now take $u = -2.07$ constant for all, and then take $\alpha=0$. Why? Earlier you said that α must be positive (line 115). Why is u constant for all, and if you take $\alpha=b=0$, then you don't have a double-well potential anymore but a simple damped harmonic oscillator. This jump isn't clear to me at all.

Thank you for your question. The previous procedure has been revised, and the estimation of γ was conducted based on the characteristic time of the GFR dynamics from the non-dimensionalization of Eq. S.1., rendering the α , b and λ parameters unnecessary.

4.6c.10. Then you assume that $\lambda = 1$ and get an estimate for γ . However, here also, it isn't clear why you assume that $\lambda=1$. You then use this value of γ for all patients, however, it isn't clear at all why it would hold for different patients considering the unjustified assumptions you took along the way.

Many thanks for this question. As we mentioned in the previous question (4.6b.10), we made the system dimensionless and the γ was estimated from the characteristic time of equilibrium without using α , b , and λ parameters.

4.7.10. Lines 331-333 – you write that the first-year GFR value is a good proxy of the functional graft performance, however based on the sigmoidal regressions you showed in S5-8, it isn't entirely convincing.

Thank you very much for your insightful comment. We appreciate your critical observation, however this statement is based on the overall clinical experience and prior studies discussed in our introduction. Based on the real data spanning 15 years of graft performance and annual GFR values, we observed a sigmoidal behavior. However, we acknowledge that the fitting is rather noisy and one cannot pinpoint an exact inflection point (GFR threshold). In light of this uncertainty, we selected a range of critical GFR thresholds (30-50 mL/min/1.73 m²) for use in the uncertainty analysis later in the study. This approach allows us to account for the variability and provides a more robust framework for predicting graft performance.

We mentioned that in the text after S5-8 figures:

“Based on real data spanning 15 years of graft performance and annual GFR values, sigmoidal behavior is observed. However, the fitting is rather noisy, making it difficult to pinpoint an exact inflection point (GFR threshold). In light of this uncertainty, a range of critical GFR thresholds (30–50 mL/min/1.73 m²) was selected for use in the uncertainty analysis later in the study. This approach accounts for variability and provides a more robust framework for predicting graft performance.”.

4.8.10. Fig 5. Should be much clearer. Which model parameters are estimated using GFR_365 days? Which critical thresholds and uncertain parameters are estimated? The uncertainty analysis step also isn't clear.

Many thanks for your comment. According to the feedback from the first reviewer, we changed the font to white for better visibility. We have clarified which model parameters are estimated using GFR_365 days, specified the critical thresholds and uncertain parameters, and detailed the uncertainty analysis step by indicating the changes in each parameter individually. More details regarding these steps have been included in the “Materials and Methods” section.

Therefore, we added in the last paragraph of section 2.1. the following:

“Our three-step strategy is outlined in Figure 1. The first part involves multilinear regression to predict the GFR (glomerular filtration rate) after one year (Machine Learning). Cross-validation was applied to the multilinear regression, to mitigate the risk of sample bias that can occur with a single partitioning of data. The second part focuses on personalized dynamic model parameter estimation for each patient based on the regression results (Mathematical Modeling). The final part deals with identifying the sources and magnitude of model uncertainty analysis and the classification of patients according to their GFR threshold regarding graft survival (Monte-Carlo).”.

4.9.10. Is Fig.6 real GFR data? Where is it taken from?

Thank you very much for your comment. Figure 6 illustrates three distinct post-transplantation GFR behaviors— increasing, decreasing, and stable—based on real patient measurements selected from our dataset.

We updated the first paragraph of section 4.3. with the following text:

“Figure 6 illustrates three distinct post-transplantation GFR behaviors—increasing, decreasing, and stable—based on real patient measurements. In individual patients, the GFR dynamics may exhibit weak and damped oscillations toward a specific GFR value or show a decreasing trend that could eventually lead to graft failure.”

4.10.10. Lines 462-466 – it isn’t clear at all how λ was estimated for each individual. Especially, as stated above, you use an assumption that $\lambda=1$ to get the numerical value of γ . So how are you now estimating it? How did you reach the distribution in Fig. S9? Also, you show in that figure that most values of λ fall within [0-0.2], so how could you assume that $\lambda=1$, use it to find γ , and then use the value of γ to give a much lower value of λ ? The whole treatment of λ is very unclear.

Thank you very much for your comment. Regarding the first part of this question, we used a different approach of estimating γ without any use of α , b , γ , and λ (please read our response in your comment 4.6b.10). Since we have three parameters for each patient α , b , an average γ , and θ —the particle swarm parameter estimator was employed to determine the value of λ from individual data from the patients without a rejection. This was done using the best GFR value obtained six weeks post-transplantation and the GFR measurements over a year. After estimating the GFR for 354 patients, the distribution shown in Figure S.9 was obtained. The observation that λ values mostly range below 0.2, with some at one, indicates that for some patients, the GFR varied initially, resulting in smaller λ values, while for others, the GFR remained fairly constant, leading to λ being equal to one. Finally, we have set $\lambda=1$ as the maximum value and therefore we observe a second mode around it.

Major changes

In our previous analysis, we considered the number of rejection episodes occurring throughout the entire study period for each patient. In the current analysis, we refined this approach by focusing on the number of rejections within the first year post-transplant. This adjustment introduces additional uncertainty into our methodology, allowing us to better assess the robustness and efficiency of our analysis. Consequently, the entire analysis was repeated from the beginning, and while the results were updated, the overall objectives and conclusions of the study remain unchanged.

The authors would like to extend their sincere gratitude to the reviewers for their thorough evaluation and insightful comments, which greatly improved the quality of this article. Their expertise and constructive feedback have been invaluable in shaping the final manuscript.

Dear Editor,

Thank you for your valuable feedback on our manuscript. We would like to express our sincere gratitude for the time and effort you invested in reviewing our manuscript, " Identification of a Personalized eGFR Threshold Improves the Prediction of Kidney Failure Risk After Transplantation." submitted to Communications Medicine Journal (COMMSMED-23-0937-T). The reviewers' constructive feedback and insightful comments have been instrumental in enhancing the quality and clarity of the manuscript. We have addressed all these comments and both a clean and marked-up copy of the manuscript with supporting information have been submitted along with this review. Please note that the reviewer's comments are given in black bold font and our replies are in blue. The structure of this report is given in the following content section:

Contents

Reviewer 1	2
Question:.....	2
Reviewer 3:	2
Question:.....	2
Reviewer 4:	4
Question 1:.....	4
Question 2:.....	5
Reviewer 5:	6
Question 1:.....	6
Question 2:.....	7
Question 3:.....	8
Question 4:.....	8
Question 5:.....	9

Reviewer 1**Question:**

Overall I'm satisfied by the edits and authors responses to reviews, from the perspective of a reviewer with Machine Learning expertise. I would note that it might be made clearer on page 4 that while cross validation was applied to multilinear regression it was not applied to the later part, perhaps with some additional context provided for clarity as to why.

Also in the discussion section I would strongly encourage the authors to lay out any potential advantages their approach may have had (with respect to differences in how the analysis was conducted) in contrast with the other ML analyses, and how this may have impacted the performance differences observed (even if it alludes to future work that should be examined).

Response:

Thank you very much for your thoughtful comment. We have revised the text on page 4 to provide a clearer explanation as to why cross-validation was applied only to the multilinear regression component and not to the subsequent stages (Mathematical Modeling and Monte Carlo simulation). The updated text, which appears on page 5, now offers additional context to justify this distinction and to clarify the methodological rationale. Therefore, the new text on page 5 is “*Cross-validation was applied to the multilinear regression step to ... quantification over cross-cohort generalization.*”

Also, we have updated the Discussion section to include a new paragraph on page 17 that outlines the key advantages of our approach in comparison with previous machine learning approaches in the literature. Thus, the new text on page 17 is: “*This framework is designed to accurately predict ... solution of optimal control problems^{21,22}.*”

Reviewer 3:**Question:**

The state of then art in nephrology and robust modeling longitudinal process jointly with time-to-event are still missing. The authors seemed to propose an home-made methodology. They have to demonstrate its performance against the state of the art, for instance by simulations. Such an application does not allow us to validate the proposal. I believe that this paper should not be considered.

Response:

This proposes a dynamic mechanistic model, in the form of Ordinary Differential Equation (ODE), to accurately predict graft function (GFR values) and transplant failure probabilities, using minimal pre-transplant data and early post-transplantation GFR measurements (only 2). Joint statistical models and dynamic mechanistic models (ODE) both deal with time-evolving data, but they do so from fundamentally different perspectives:

- Joint models are statistical, and they describe empirical correlations between longitudinal variable and event risk without explicit mechanistic assumptions about the underlying biology.
- Dynamic mechanistic models may use ODEs (or PDEs, stochastic models, agent-based models) to describe causal processes based on biophysical, biochemical, or physiological mechanisms.

The choice of a dynamic model and not a statistical one is based on the following:

- (i) *Limited time points*: we want to make predictions using very few time points (two). There is no statistical process model that can perform accurately in this condition.
- (ii) *Modeling interventions*: Joint models are data-driven and therefore extrapolating beyond observed covariate values and biomarker trajectories is risky. ODEs can extrapolate to new regimes (within limits of model validity) — e.g., simulate dosing changes, drug effects, or extreme scenarios. Since in the future we aim to model therapies, this justifies our choice.
- (iii) *Non-linear behavior of the GFR values*: The data exhibit a non-linear behavior (existence of two states and oscillations) that can be naturally captured by the selected model. The typical joint model uses mixed effects and survival that are linear trends with random variation.
- (iv) *Extensions and biological interpretability*: Statistical joint models do not capture the underlying biological phenomena. Also, adding a relevant mechanism, e.g., a chronic infection, is not feasible. In the ODE context, we can easily model additional biological phenomena.
- (v) *Predicting time to failure*: This is a core feature of joint and survival models. However, having an individually parametrized equation, we could also predict this. Here we chose to work towards a classification problem rather than predicting the exact time of failure.

Although in this paper we deal with the problem of failure prediction, our framework unlocks the toolbox of control theory. The latter allows us to include the improved initial predictions after follow-up GFR measurements (Bayesian filtering) or find optimal therapies (optimal control problem). Moreover, we can include stochastic uncertainty and use ideas from stochastic differential equations. Therefore, the selection for our “home-made” algorithm is much deeper and extendable than a mere statistical/ML prediction method. This justification is now clearly stated in the manuscript. More precisely, we have added the following as the fourth paragraph in the Discussion section:

“This framework is designed to accurately predict graft function, as reflected by GFR values, and ... solution of optimal control problems^{21,22}.”

The new text provides a detailed justification for employing a dynamic mechanistic model (ODE-based), distinguishing it from joint statistical models, and outlining the advantages of our approach, particularly in terms of interpretability, extrapolation, and future extensibility.

Reviewer 4:

I thank the authors for their revisions and clarifications regarding the Duffing oscillator–based approach for GFR evolution. The updated manuscript appears considerably improved. I have several specific comments regarding the manuscript's treatment of the Duffing oscillator model for GFR evolution:

Question 1:

The handling of γ (gamma): Is a constant γ value used across all patients in the simulations? Does Table S4 only include data from patients without rejection? If the γ value from non-rejection patients is applied to model potential rejection cases, what is the justification? Has sensitivity analysis been performed to demonstrate result robustness across different γ values? What is the expected biological variation in γ between rejection/non-rejection cases? How critical is γ to the model outcomes? The revised treatment of λ (lambda) is now more rigorous and well-justified.

Response:

Thank you for your insightful question regarding the treatment of the parameter γ in our model. We are pleased to clarify the reasoning and methodology behind our approach:

Yes, a constant γ value was used across all patient simulations. Specifically, the nominal value of $\gamma=0.000032 \text{ d}^{-2}$, estimated from patients without rejection, was applied consistently in both rejection and non-rejection cases. Table S4 indeed includes data exclusively from the non-rejection cohort, which was used to calibrate the baseline parameters of the model. To justify this decision, we conducted a global sensitivity analysis using the RS-HDMR method, as described in the Supplementary Information. This analysis quantitatively assessed the relative influence of four model parameters— α , b , γ , and λ —on GFR dynamics. The results demonstrated that γ had the lowest first- and second-order sensitivity indices (0.021 ± 0.007), indicating that variations in γ have a minimal impact on the model's output compared to the other parameters. Consequently, we held γ constant in all patient-specific estimations to reduce the dimensionality of the estimation problem and focus on more influential parameters. While we acknowledge that some biological variability in γ may exist between rejection and non-rejection cases, the low sensitivity supports the robustness of the model's predictions to such variations. Moreover, due to the limited availability of reliable early post-transplant data in the rejection cohort, estimating γ individually would risk overfitting and reduce the generalizability of the model. We believe this strategy offers a balanced trade-off between biological realism and model parsimony, allowing us to preserve predictive performance while maintaining interpretability. Future work involving longitudinal GFR data from a larger rejection cohort could enable a more personalized estimation of γ and further validate these findings. Table S4 presents data from all patient subgroups in the training cohort (Table 1), including patients with no rejection, one rejection, and multiple rejections. The title of Table S4 was changed to “*Table S. 4: Average loss per year in the whole training cohort of patients (Table 1).*”

Therefore, in the main manuscript, we updated on page 22 the paragraph: “*With the individualized ... based on their clinical GFR measurements.*”. Also, in SI2, we added the following paragraph in section

4 (page 10): “Before proceeding with the estimation ... estimated using a Particle Swarm Optimization (PSO) approach.”.

Question 2:

Visualization Recommendation

I strongly recommend adding a comprehensive schematic, graphic figure (ideally as Figure 1) that illustrates:

The complete patient workflow

Step-by-step methodology

Key decision points and processes

Example patient progression through the model

This visual pipeline would significantly enhance reader understanding of the methodology and make the paper more accessible to a broader audience. This figure should illustrate the actual steps as currently written in Figure 1.

Response:

Thank you for this valuable suggestion. In response, we have added a comprehensive schematic labeled as Figure 1 to visually convey the methodological framework of our study. The new figure is the following:

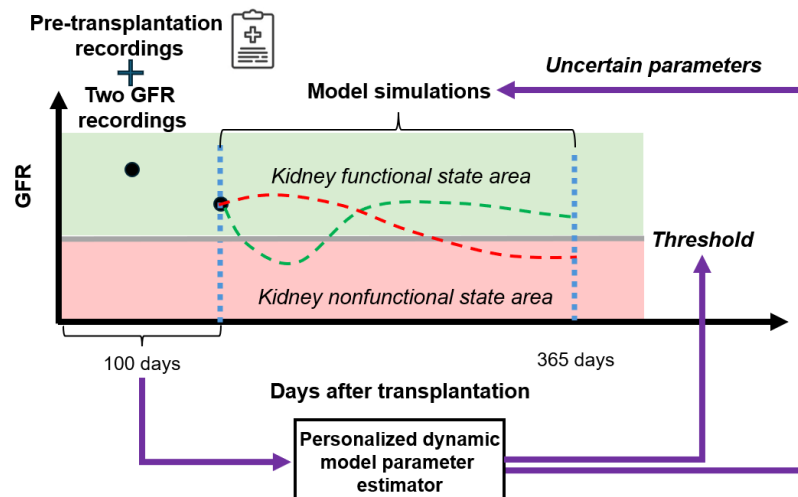


Figure 1: Schematic of model simulations for kidney graft functional and nonfunctional trajectories

This figure was carefully designed to illustrate the full patient workflow, starting from pre-transplantation data and early post-transplant GFR recordings through to long-term graft outcome prediction. It provides a step-by-step representation of the methodology, including first-year GFR trajectory generation, personalized parameter estimation, and integration into the dynamic model. To enhance clarity, the figure also highlights the distinction between functional and nonfunctional kidney state areas relative to the clinical threshold, together with representative simulated trajectories. This allows readers to readily visualize how individual patient data are incorporated into the model and how uncertainty in the estimated parameters propagates into long-term graft failure prediction. Accordingly, we updated the final paragraph of the Introduction—“*After reviewing existing modeling ... the materials and methods are described in Section 4.*”, and added Figure 1 to enhance clarity and accessibility.

We believe this visual representation significantly enhances the clarity of our approach and makes the methodology more accessible to a broader readership, as you recommended.

Reviewer 5:

In this manuscript, the authors attempt to prove a personalised eGFR threshold to detect post-transplant kidney failure risk.

Question 1:

You mention the analysis indicates that an eGFR threshold exists that allows for classifying high-risk patients. I struggle to accept there is a sharp dichotomised eGFR threshold for risk above or below with risk for kidney failure risk dramatically changes. eGFR and risk for kidney transplant failure is a continuous variable and influenced by many other variables such as proteinuria. I cannot see the latter factored in at all? A high eGFR but high proteinuria would not reassure me of low risk for graft loss.

Response:

We would like to thank the reviewer for allowing us to clarify this point further. Our intention was to examine the eGFR as the principal input of our model. The existence of two steady states (failed and successful) implies the existence of a GFR threshold (saddle node). Although the deterministic model assumes a sharp threshold, we have discovered that there is a range of values (30-50mL/min/1.73 m²) that quantify the uncertainty about the critical GFR threshold. Real data spanning 15 years of graft performance and annual GFR values guided this approach, as shown in Supplementary Figures S.5 to S.8. A sigmoidal trend can be seen in these graphs, which show the graft survival chance as a function of GFR for different patient groups. But because of noise in the data, it's still not clear where the exact turning point is. We can therefore confidently assume that there is a range of GFR critical values. In our analysis we embrace this uncertainty (and the λ parameter), and this is the reason our deterministic model provides probabilistic results. The latter implies that even in our model patient's eGFR crosses

the theoretical threshold, it is not deterministic that the graft will fail or be successful. Of course, if we would like further constrain this uncertainty, we should include further factors in our analysis as suggested by the reviewer.

Notably, a previous analysis from Yoo et al. support our approach and results. Different types of machine learning were used in the study (DOI:10.1038/s41598-017-08008-8), and a serum creatinine cutoff of 1.65 mg/dL was found. The threshold wasn't chosen at random; it was set using the CART (Classification and Regression Trees) algorithm and data. This algorithm checks possible split points using factors like the Gini impurity to make predictions as accurate as possible. In their model (illustrated in Figure 2B of the publication), serum creatinine level appeared at the first decision node, signifying its dominant role in stratifying patients according to 10-year graft outcomes. Clinically, this threshold corresponds approximately to an eGFR of 30–40 mL/min/1.73 m², closely matching the critical range identified in our survival model and reinforcing the biological relevance of this threshold in predicting graft failure risk. Graft failure rates were much greater in patients with creatinine levels exceeding this threshold.

We have updated Subsection 2.3 by changing its title from “A GFR threshold exists with respect to graft survival” to “A GFR threshold range exists with respect to graft survival”, and by adding the following text at the end: *by Yoo et al.¹⁸, ... predicting graft failure risk.*” which contextualizes our findings in light of previous work by Yoo et al. This addition highlights the alignment between our data-driven critical GFR threshold and the serum creatinine cut-off identified in their decision tree analysis, thereby reinforcing the clinical relevance of our selected threshold range.

Question 2:

On page 14 you mention the generalizability of your approach with sensitivity analysis stratifying both the training and validation cohorts based on patient sex, donor sex, and patient age. Why did you choose these variables - especially donor sex? That seems an odd choice to show generalisability. Generalisability should be in a different ethnic groups, different immunosuppression regimens etc.

Response:

As this is a monocentric study cohort, the sensitivity analysis could only concentrate on available subgroups. E.g., other ethnic groups were scarce (<1%) in our cohort and immunosuppressive regimen were quite uniform. On the other hand, different age groups and patient sex are typical subgroups which are primarily considered in many studies (transplant and non-transplant) and sensitivity analyses.

To address the limitation concerning generalizability across different ethnic groups and immunosuppressive regimens, we have updated the sixth paragraph of the Discussion section by adding the following text:

“Our approach has several limitations. The data used for model development and validation were collected from a single center, primarily serving patients of Caucasian origin, with other ethnic groups

being scarce (<1%). The immunosuppressive regimens administered across the cohort were also quite uniform.”

Regarding donor sex, a lower rate of delayed graft function with female donor organs was reported from an analysis of 25,140 kidney graft recipients from the British transplant registry (Morgan G et al, Transplantation 2020, 104:1033-1040). Therefore, we do see sensitivity analyses for sex justified to confirm applicability of the model to these important subgroups. Therefore, we do see sensitivity analyses for sex justified to confirm applicability of the model to these important subgroups.

Question 3:

I struggle to understand the clinical application of this work. Graft function and the performance of the kidney allograft is based on many donor, recipient and surgical factors and numerous post-transplant variables.

Response:

We agree that many pre-, peri-, and posttransplant factors modify graft function and survival. These are—in part—represented by the early changes in GFR, which were used in our modeling. Our approach is intended as a generic framework, which could indeed be enriched by additional variables in the future. The great advantage of our framework is its extendibility and integration of further tools and features. Our intention is to turn this framework into a clinical tool.

We have revised the abstract to include the following sentence:

“The aim of this paper is to present this foundational model and highlight its potential to evolve into a clinical tool for personalized patient management.”

This addition makes our approach even more general, showing that it could be used in a wide range of clinical settings to help individuals make decisions.

Question 4:

I am not sure why there is such focus on stratification by rejection but nothing for other causes for graft dysfunction.

Response:

Rejection was taken as a main example since it is a major determinant of graft dysfunction and that rejection was present in a substantial number of patients.

Other causes of graft dysfunction, such as BK virus nephropathy, was far lower prevalent (6.4%, n=113).

Question 5:

How would this compare with tools like iBox? What would be its advantages or disadvantages?

Response:

The iBox attempts a prediction in a different setting, starting with several transplant variables and including detailed biopsy findings. Using such broad approach, the c index was approximately 0.80. Our intention was different; to develop a data-driven model that relies solely on early changes in GFR. Of course, this approach is less accurate regarding future predictions, yet, it is a generic framework that can be enriched by additional variables.

The authors would like to extend their sincere gratitude to the reviewers for their thorough evaluation and insightful comments, which greatly improved the quality of this article. Their expertise and constructive feedback have been invaluable in shaping the final manuscript.

Best regards,
Haralampos Hatzikirou

Associate Professor
Department of Mathematics,
Khalifa University
IP.O. Box 127788, Abu Dhabi, UAE
E-Mail: haralampos.hatzikirou@ku.ac.ae

Rebuttal

The core concern—lack of robust comparison with existing methods—is addressed on two fronts. First, the submitted manuscript already benchmarked our framework against state-of-the-art classifiers (XGBoost, Random Forest), where our integrated regression–dynamical–uncertainty approach performed better (e.g., RF/XGBoost F1 ≈ 0.74 – 0.75 vs ours ≥ 0.81 at 1 year).

Second, we added three linear mixed-model (LMM) baselines (LMM1–3) and show our endpoint regression is equal or superior across patient strata in predicting the 1st year GFR (e.g., one-rejection: R^2 0.600 vs 0.588; multiple-rejection: RMSE 16.99 vs 17.88–20.84).

Beyond accuracy, our mechanistic, uncertainty-quantified model is justified by minimal early measurements, nonlinear dynamics, and prospective interpretability for therapy modeling. Predictions are externally validated across a temporally independent cohort (AUC ≈ 0.90 ; F1 ≈ 0.83 at 1 year) and remain robust at 2 years.

This appeal addresses the comments of Reviewer #3, as stated below in two consecutive rounds and final editorial decision:

Reviewer #3 (Remarks to the Author):

The state of then art in nephrology and robust modeling longitudinal process jointly with time-to-event are still missing. The authors seemed to propose an home-made methodology. They have to demonstrate its performance against the state of the art, for instance by simulations. Such an application does not allow us to validate the proposal. I believe that this paper should not be considered.

Here, we used three different linear mixed models. Linear mixed models (LMMs) extend linear regression to data with grouped or repeated measurements (e.g., patients measured over time). They include fixed effects for population-level relationships and random effects for subject-specific deviations, which properly account for within-group correlation and allow partial pooling [1,2].

The first lineal mixed model (LMM1) uses the GFR value at 42 days and captures the outcome for patient i at 365 days. Its equation is the following:

$$y_{it} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i}) \cdot Time_{c,it} + \varepsilon_{it}$$

Where the centered $Time_{c,it}$ equals:

$$Time_{c,i365} = 1 - \frac{1}{2} \left(\frac{42}{365} + 1 \right) = 0.4424$$

u_{0i} and u_{1i} are equal to:

$$\begin{bmatrix} u_{0i} \\ u_{1i} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{bmatrix} \right)$$

and ε_{it} is equal to:

$$\varepsilon_{it} \sim N(0, \sigma^2)$$

τ_0 , τ_1 , and τ_{01} are equal to :

$$\begin{aligned} \tau_0^2 &= Var(u_{0i}) \\ \tau_1^2 &= Var(u_{1i}) \\ \tau_{01} &= Cov(u_{0i}, u_{1i}) \end{aligned}$$

The second linear mixed model (LMM2) uses GFR values at 42 and 100 days, along with baseline covariates, to capture the outcome for patient i at 365 days. Its equation is the following:

$$y_{it} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i}) \cdot Time_{c,it}$$

$$+ \sum_k \gamma_k x_{ik} + \sum_k \delta_k Time_{c,it} x_{ik} + a_1 y_{i42,t=365} + a_2 y_{i100,t=365} + \varepsilon_{it}$$

Where x_{ik} are the baseline predictors, $Time_{c,it} x_{ik}$ are included when time interactions are enabled, and a_1 and a_2 are active at time $t=365$ days.

Lastly, the third linear mixed model (LMM3) uses the values of GFR at 42 and the percentage of GFR change between 42 and 100 days, and baseline covariates to capture the outcome for patient i at time 365 days. Its equation is the following:

$$y_{it} = (\beta_0 + u_{0i}) + (\beta_1 + u_{1i}) \cdot Time_{c,it}$$

$$+ \sum_k \gamma_k x_{ik} + \sum_k \delta_k Time_{c,it} x_{ik} + a_1 y_{i42,t=365} + a_2 perc_{i100,t=365} + \varepsilon_{it}$$

Where x_{ik} are the baseline predictors, $Time_{c,it} x_{ik}$ are included when time interactions are enabled, and a_1 and a_2 are active at time $t=365$ days.

The non-rejection patients in the training cohort were used to evaluate the model's performance using 5-fold cross-validation. Table 1 shows the performances of the three LMMs in non-rejection, one rejection, and multiple rejection patients, and also contains the performance of **our end-point regression model** with feature selection.

Table 1: Comparisons of LMM models with our model used in the manuscript for the non-rejection patients

	R² range	RMSE range
LMM1	0.353 – 0.672	11.310 – 15.173
LMM2	0.579 – 0.678	11.240 – 14.540
LMM3	0.560 – 0.704	11.748 – 13.591
Regression	0.660	12.540

As for the one rejection patients, the results are provided in Table 2.

Table 2: Comparisons of LMM models with our model used in the manuscript for the one-rejection patients

	R²	RMSE
LMM1	0.432	17.082
LMM2	0.589	15.057
LMM3	0.541	15.901
Regression	0.600	14.070

As for the multiple rejection patients, the results are provided in Table 3.

Table 3: Comparisons of LMM models with our model used in the manuscript for the multiple-rejection patients

	R²	RMSE
LMM1	0.320	18.787
LMM2	0.483	17.938
LMM3	0.408	19.190
Regression	0.460	16.990

Across identical patient-grouped cross-validation splits, LMM1—the simplest mixed model—performed worst (lowest R², highest RMSE). In contrast, LMM2 and LMM3 achieved R² and RMSE values that were very close to those of our initial model, with only small differences. These results indicate that the linear mixed-model formulations did not improve 1-year eGFR

prediction over the initial model, and therefore, we retain the initial model as a useful and parsimonious choice.

References

- [1] L. Fahrmeir, T. Kneib, S. Lang, B.D. Marx, Regression, Springer Berlin Heidelberg, Berlin, Heidelberg, 2021. <https://doi.org/10.1007/978-3-662-63882-8>.
- [2] Y.-C. Lin, T.-S. Lai, S.-L. Lin, Y.-M. Chen, T.-S. Chu, Y.-K. Tu, The impact of baseline glomerular filtration rate on subsequent changes of glomerular filtration rate in patients with chronic kidney disease, Sci Rep 11 (2021) 7894. <https://doi.org/10.1038/s41598-021-86955-z>.

Additions in the manuscript and SI

The following parts were added in Section 2.2, and the second paragraph of the Discussion section, respectively:

“To provide a conventional statistical baseline, we also fitted three linear mixed-effects models (LMMs) that include: (i) a 1-year GFR model using the 42-day GFR measurement only (Eqs. S.1-S.7), (ii) a 1-year GFR model using the 42-day and 100-day measurements plus baseline covariates (Eqs. S.8-S.14), and (iii) the same model with interaction terms (Eqs. S.15-S.21)16,17. All LMMs were trained on the non-rejection training cohort and evaluated via cross-validation and on the temporally distinct validation cohort (patients with none, one, or multiple rejections). Across all comparisons (e.g., RMSE/MAE for 1-year GFR and failure-risk discrimination), the LMMs did not yield better results than our patient-specific dynamic modeling workflow; performance was comparable or lower. The regression analysis results are shown in Tables S. 3-S. 5 of Supplementary Material 2. ”

“We also explored classical longitudinal approaches based on linear mixed-effects models. All models were trained on the N1 non-rejection cohort and evaluated with the same splits and on N2. Across endpoints, the LMMs did not improve prediction of first-year GFR (higher RMSE, lower R^2) at 365 days relative to our proposed framework. This is consistent with the limited number of early post-transplant time points—where mixed models offer little advantage over simple regression—and with the fact that standard LMMs are essentially linear and therefore unable to capture the behavior of graft function that our dynamical-systems model was designed to represent. In our setting, increasing LMM complexity tended to add variance without commensurate gains in accuracy.”

In Supplementary Material 2 (SI_2.docx), we expanded Section 1 by adding the results of the linear mixed models, presented in Tables S.3, S.4, and S.5.

Any other comments

- All data, code, and methodological details necessary to reproduce results are provided with the manuscript and supplementary materials; de-identified raw data and code are available from the corresponding authors (link provided in the manuscript).
- We are happy to incorporate the LMM analyses into the revised main text or supplement, and to add any additional baseline the editor deems essential for readers.

Dear Editor,

Thank you for your valuable feedback on our manuscript. We would like to express our sincere gratitude for the time and effort you invested in reviewing our manuscript, " Identification of a Personalized eGFR Threshold Improves the Prediction of Kidney Failure Risk After Transplantation." submitted to Communications Medicine Journal (COMMSMED-23-0937-T). The reviewer's constructive feedback and insightful comments have been instrumental in enhancing the quality and clarity of the manuscript. We have addressed all these comments and both a clean and marked-up copy of the manuscript with supporting information have been submitted along with this review. Please note that the reviewer's comments are given in black bold font and our replies are in blue. The structure of this report is given in the following content section

Comment 1: Main manuscript, line 172: The abbreviation GFR has already been introduced in line 50.

Response:

We thank the reviewer for this observation. We have removed the redundant redefinition of the abbreviation "GFR" at line 172, as it is already introduced at line 50.

Comment 2: Supplementary material, line 283: The sentence appears incomplete or lacks appropriate punctuation.

Response:

We thank the reviewer for noting this. The sentence in the Supplementary Material (line 283) has been revised for completeness and punctuation, and now reads clearly in the updated manuscript. We have revised it to read:

"High-risk patients were those whose annual GFR value was below a specified threshold, with thresholds of 30, 40, and 50 mL/min/1.73 m² obtained from Section 5 of SI."